

Monitoring Cues, Peer Communication, and Coordination Among LLM Agents

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June 2026

Abstract

Language models increasingly write and relay messages between people. This paper asks whether an authority-monitoring cue shown only to LLM message writers changes what they write enough to move the decisions of LLM readers. The setting is a controlled narrative coordination task. Each agent receives a noisy private briefing about a simulated regime, exchanges short messages with peers, and then decides whether to join an uprising. Across 7 hosted model snapshots, join rates fall as regime strength rises, matching the direction of an external regime-change benchmark. The main treatment shows an authority-monitoring cue to agents only while they write messages; private briefings and receiver prompts never change. Joining falls in a full-scale primary run, and by 8.0 percentage points in a nested 500-cell Llama 3.3 70B replication. Replaying edited messages to fresh receivers isolates the text features that carry the effect. A statement that others are willing to act raises joining among valid responses by +7.94 points in direct language and by +5.50 points in coded language. Valid-response rates are balanced across these arms, making willingness content the strongest mechanism result. A statement about repression risk lowers joining among valid responses by 3.99 points, but valid-response rates differ across those arms, so that contrast is suggestive rather than dispositive. A control that makes messages coded without mentioning monitoring barely moves participation. In the tested snapshots, monitoring cues change what senders write about willingness and risk, and those changes are enough to move otherwise matched receivers.

Short abstract (100 words). An authority-monitoring cue shown only to LLM message writers changes their messages and lowers downstream LLM participation, while private briefings and receiver prompts stay fixed. A nested 500-cell Llama replication shows an 8.0-point decline. Replaying edited messages isolates the main pathway: stating that others are willing to act raises joining among valid responses by +5.50–+7.94 points. Risk language is a second pathway, more sensitive to missing responses. Induced coded style alone has little effect. The estimates describe tested model snapshots in a simulated task.

JEL: C72, C92, D82, D83, P16

Keywords: monitoring cues, peer communication, coordination, LLM agents, sender adaptation, message replay, coded speech

1 Introduction

Monitoring can alter coordination before decision makers ever see a warning, because it can change what their peers say. This paper asks a narrow causal question about large language model (LLM) behavior. Suppose an authority-monitoring cue is shown only to the LLM agents who write messages. The simulated state, each private briefing, the receiver model, and the receiver’s decision prompt are all held fixed. Do the messages change enough to move downstream LLM participation? In the tested environment, they do. Joining falls in two full-scale model families. Interventions on the message text then show that statements about others’ willingness to act are the strongest experimentally manipulated carrier of the effect.

The question connects two literatures, without treating the simulation as evidence about human behavior. Research on authoritarian control shows that censorship and surveillance can suppress expression and collective action (King et al., 2013; Xu, 2021; Penney, 2016; Stoycheff, 2016). In observed data, however, several forces move together: senders change what they say, platforms delete content, samples are selected, and receivers are directly afraid. At the same time, language models are increasingly used to draft, summarize, and relay messages. If message generation is sensitive to monitoring cues, that is a fact about model behavior and a question for system evaluation.

The experiment uses a regime-change coordination problem as a disciplined setting. A latent regime strength generates noisy private signals. Each signal is rendered as a natural-language briefing, and each agent chooses whether to join an uprising. Actions are strategic complements: joining pays off only if enough others join. That makes peer messages potentially consequential, because a message can describe the regime and can state whether others appear ready to act. The canonical Morris–Shin attack-mass curve supplies an external benchmark for the direction of the response (Carlsson and van Damme, 1993; Morris and Shin, 2003). It is not an equilibrium test of the implemented agents, who are never shown the prior, the payoff parameters, the group size, or the success threshold.

The closest designs leave a specific gap. Avoyan (2024) studies pre-play communication in human global games but does not vary monitoring at the sender stage. Stoycheff (2016) studies a surveillance prime and willingness to express political views but does not observe a full sender-message-receiver chain. King et al. (2013) distinguishes criticism from collective-action content in censored field data. Edmond (2013) models a regime that manipulates citizen information. Existing designs rarely combine four elements: monitoring randomized on the sender side only, observation of every message, fixed receiver inputs, and replay interventions on the generated text.

I implement that combination with hosted LLM snapshots. Across 7 models, each agent receives a private prose briefing, exchanges messages with neighbors on a fixed network, and then chooses JOIN or STAY. The main treatment adds an authority-monitoring warning to the message-writing prompt and nowhere else. The receiver never sees the warning. The

receiver’s private briefing and decision prompt are bit-identical across treatment arms. What the experiment estimates is therefore a response of the tested model snapshots and decoding configuration. It is not an estimate of effects on citizens, regimes, or welfare.

Three findings summarize the experiment. First, join rates decrease monotonically with regime strength in all seven models (mean correlation with the benchmark attack mass, $r(J, A(\theta)) = +0.80$). The prose environment therefore produces the intended directional response to the state. Second, allowing communication has effects that differ across models and average near zero when each model receives equal weight. Third, the monitoring cue lowers joining by 13.9 percentage points in the full-scale primary run and by 8.0 points in a nested 500-cell Llama 3.3 70B replication. A control that makes messages more metaphorical without mentioning monitoring moves joining by only -0.7 points.

Fixed-message interventions sharpen the mechanism. In these experiments, receivers see edited versions of real messages instead of live peer messages. Some receiver responses cannot be parsed into a JOIN or STAY decision; I call the join rate among the responses that can be parsed *parseable joining*. Holding the factual regime assessment fixed, adding an explicit claim that others are willing to act raises parseable joining by +7.94 points in direct language and by +5.50 points in coded language. Valid-response rates are nearly identical across these arms. The contrasts also stay positive when every unparseable response is recoded uniformly as JOIN, or uniformly as STAY. A rewrite that only adds repression-risk language lowers parseable joining by 3.99 points. But valid-response rates differ across the risk arms, and one of the two uniform recodings erases that contrast. The evidence therefore supports willingness content as the primary experimentally manipulated pathway, and risk language as a plausible but less secure second pathway.

The contribution is to take a sender-mediated model response apart experimentally. The design changes only the sender prompt, observes every message that change produces, and measures the effect of those messages on matched receivers. Replay interventions then vary specific message features one at a time. Section 2 positions the paper. Sections 3 and 4 define the benchmark and the design. Section 5 reports the monitoring and replay results. Section 6 concludes.

2 Related Literature

The regime-change framework supplies a strategic-complementarity benchmark; it is not the paper’s outcome of interest. Morris and Shin (2002) study the welfare consequences of technically public information, and Cornand and Heinemann (2014) measure how subjects weight private and public signals. The present design varies none of those objects: no common signal, no common knowledge, no welfare calculation. Its closest coordination comparator is Avoyan (2024), who studies pre-play communication in global games theoretically and with human subjects. Relative to that design, this paper uses unrestricted natural language, randomizes an authority-monitoring cue only at the moment senders write, observes every generated message, and replays manipulated messages to matched receivers. The global-game tradition from Carlsson and van Damme (1993) through Morris and Shin (2003), the dynamic extensions of Angeletos et al. (2007) and Chassang (2010), and classic cheap-talk results (Crawford and Sobel, 1982; Farrell and Rabin, 1996; Blume and Ortmann, 2007; Ellingsen

and Östling, 2010) motivate the coordination setting. Edmond (2013) is the closest formal comparison. His regime changes the citizens’ signal environment; the treatment here changes the language that citizens generate for one another.

Authoritarian control works through both information and coordination. Kuran (1991) models preference falsification under social pressure. Shadmehr and Bernhardt (2011) characterize collective action under repression. Guriev and Treisman (2019) describe informational autocrats, and Little (2016) models communication technology in protest. On the empirical side, King et al. (2013) show that Chinese censorship targets collective-action content, Xu (2021) studies digital surveillance as preemptive control, and Penney (2016) and Stoycheff (2016) examine surveillance announcements and online expression. These studies motivate the paper’s central distinction—monitoring can deter receivers directly, or it can change what senders say—but none observes the controlled sender-message-receiver sequence studied here. Beliefs about others’ willingness to act are central in the human experiments of Bursztyn et al. (2020b), Bursztyn et al. (2020a), and Cantoni et al. (2019). The last of these finds strategic substitutability in the Hong Kong movement, a reminder that complementarity is an imposed feature of the present task rather than a general fact about protest. In the simulated environment here, direct receiver exposure to the warning is held fixed while sender-generated messages vary.

The mechanism also differs from information design, which separates the movement of beliefs from the mapping of beliefs into action. Bayesian persuasion treats information as a chosen distribution of posterior beliefs (Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019; Mathevet et al., 2020). Gans (2026) emphasizes that actions can be steered without changing beliefs, by changing the procedure that turns a belief into an action. Here the receiver’s model snapshot, decoding configuration, and decision prompt are all fixed. What the monitoring cue changes is whether the generated peer messages read as a guide to action.

Methodologically, Horton (2023) proposed studying LLMs as “homo silicus.” Aher et al. (2023) and Argyle et al. (2023) use language models to replicate human-subject results and to simulate survey samples, and Mei et al. (2024) compare chatbot behavior with human behavior in canonical games. Other work documents limits of LLM coordination (Akata et al., 2025; Sun et al., 2025); model scale does not reliably predict strategic performance (Jia et al., 2025). Most closely, Ruaño and Rajan (2026) place LLM depositors in a bank-run coordination game with optional peer posts and network contagion. The present paper adds four elements: monitoring shown only to senders, observation of every message, fixed receiver contexts, and replays that vary message features. Ouyang et al. (2024) and Cheung et al. (2025) show that alignment and prompting can shift model decisions, which makes the mapping from politically loaded prose to action an object of study in itself. The results directly characterize only the tested snapshots and prompts. Their relevance to deployed systems motivates external validation; it does not substitute for it. The concerns raised by Larooij and Törnberg (2025, 2026) motivate the falsification, classifier, and cross-model checks.

3 A Regime-Change Coordination Benchmark

A continuum of citizens indexed by $i \in [0, 1]$ simultaneously choose whether to join an uprising ($a_i = 1$) or stay home ($a_i = 0$). The regime has strength $\theta \in \mathbb{R}$, drawn from a diffuse (improper uniform) prior. States $\theta < 0$ represent regimes so weak they fall without opposition under the strict rule below; states $\theta \geq 1$ represent regimes that survive even unanimous attack. The regime falls if the mass of citizens who join exceeds θ (Eq. 1):

$$\text{Regime falls} \iff A \equiv \int_0^1 a_i di > \theta. \quad (1)$$

Payoffs depend on the citizen's action and the outcome (Eq. 2):

$$u_i(a_i, A, \theta) = \begin{cases} B & \text{if } a_i = 1 \text{ and } A > \theta \\ -C & \text{if } a_i = 1 \text{ and } A \leq \theta \\ 0 & \text{if } a_i = 0 \end{cases} \quad (2)$$

where $B > 0$ is the payoff to joining a successful uprising and $C > 0$ is the cost of joining a failed attempt. Non-participants receive zero regardless of the outcome.

Each citizen observes a private signal $x_i = \theta + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently across citizens.

Proposition 1 (Morris and Shin, 2003). *In the limit of diffuse priors, there exists a unique Bayesian Nash equilibrium (BNE) in threshold strategies. An agent joins if and only if $x_i < x^*$ (Eq. 3), where*

$$x^* = \theta^* + \sigma \Phi^{-1}(\theta^*) \quad (3)$$

$\theta^* = B/(B + C)$, and Φ denotes the standard normal CDF.

The *attack mass* $A(\theta)$, the theoretical fraction of the population that joins at regime strength θ , is given by:

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right). \quad (4)$$

This is a decreasing function of θ : weaker regimes face larger uprisings.

I define $J(\theta)$ as the *empirical join fraction*: the finite-sample, observed counterpart to the theoretical attack mass $A(\theta)$. Each experimental cell uses $n = 25$ agents. The distribution that generates θ is chosen by the researcher; agents are shown neither that distribution nor (B, C, σ) . The continuum $A(\theta)$ serves as an external behavioral benchmark throughout. Throughout the paper, $r(J, A(\theta))$ denotes the Pearson correlation between the empirical join fraction J and the benchmark attack mass $A(\theta)$ across period-level observations. This correlation is a construct check on the language environment. It verifies that agents use the prose signal monotonically before the paper turns to the surveillance estimates.

The textbook model treats the information structure as given: signals arrive, agents update, and actions follow. The experiment adds a language stage before play. Agents first turn their private briefings into messages. Other agents then turn their own briefing, plus the messages they receive, into actions. Let $\mathcal{B}_i$ denote sender i 's private briefing object: the prose

Table 1: Canonical global-game objects and their implemented analogues. The continuum equilibrium enters only as an external benchmark; the implemented environment is a controlled language-mediated coordination game, not an isomorphic transcription of Morris–Shin.

Canonical object	Implemented analogue
Fundamental θ (regime strength)	Researcher-drawn $\theta \sim \mathcal{N}(z_{ct}, 1)$; never shown to agents
Private signal $x_i = \theta + \varepsilon_i$	Same draw, rendered as a multi-paragraph natural-language briefing over eight evidence domains
Prior over θ	Diffuse in theory; agents see neither the prior nor the country–period mean z_{ct}
Payoffs (B, C)	Embedded in narrative stakes; numeric (B, C) never shown; benchmark computed at $B = C = 1$
Signal noise σ	Fixed at $\sigma = 0.3$; not shown to agents
Critical-mass success rule ($A > \theta$)	Implicit in the JOIN/STAY framing; agents are told neither the threshold nor the group size
Population (continuum $[0, 1]$)	Finite $n = 25$ agents per period-level cell
Communication structure	Pre-play messages on a Watts–Strogatz network ($k = 4, p = 0.3$)
Message generation	LLM senders map briefing $\rightarrow$ message; surveillance changes this conditional distribution $\mu(m \mid \mathcal{B}_i)$
Equilibrium object	Continuum BNE attack mass $A(\theta)$, used only as an external behavioral benchmark

briefing generated from the normalized signal, together with the briefing generator’s language context. (The full LLM context also includes the game instructions; $\mathcal{B}_i$ isolates the part held fixed across baseline and surveillance.) Let $\mu_0(m \mid \mathcal{B}_i)$ be the distribution of messages m generated from that briefing at baseline, and let $\mu_S(m \mid \mathcal{B}_i)$ be the distribution generated under monitoring. The surveillance treatment changes only this conditional distribution. It does not change θ , the private briefing, or the receiver’s decision prompt. The question is therefore whether replacing μ_0 with μ_S changes $J(\theta)$.

The replacement can matter even if the messages still carry information about θ . A peer message can contain two things at once: a factual assessment of the regime, and a statement about whether others appear willing to act. The experiment measures how the monitoring cue changes each feature, and whether the changed messages move receivers. Coded language is one observable feature of the generated messages, not the theoretical object itself. The codedness control, the decoding test, and the fixed-message interventions in Section 5 vary form and content separately.

The briefing generator is the system that converts z -scores into natural-language intelligence briefings. It uses three latent sliders—direction, clarity, and coordination—to select phrases across eight evidence domains. The result is not a transcription of the scalar Morris–Shin signal. It is a controlled language environment inspired by the canonical model, with explicit content about the state, the clarity of the evidence, and the prospects for action. Table 1 maps each canonical global-game object to its implemented analogue. Full generator details are in Appendix E.1.

4 Experimental Design and Behavioral Benchmark

The experiment has two phases. The first phase asks whether LLM join rates respond monotonically to regime strength when private signals arrive as natural language. It uses a private-information treatment, a communication treatment, and falsification tests. The second phase asks whether an authority-monitoring cue changes model-generated messages and, through them, receiver decisions. All LLM interactions use the same prompt structure across models.

In the theory (Section 3), θ is an unbounded fundamental. In the benchmark experiments, θ is drawn from a normal distribution, and agents are never shown the payoff parameters (B, C) . I compute the benchmark $A(\theta)$ under the canonical normalization $B = C = 1$, which gives $\theta^* = 0.50$, with $\sigma = 0.3$.

Some notation recurs. The theoretical fundamental cutoff is $\theta^* = B/(B + C)$, and the signal cutoff is $x^* = \theta^* + \sigma\Phi^{-1}(\theta^*)$. The empirical cutoff $\hat{\theta}^*$ is the point at which the fitted state curve crosses a 50% join rate. The fitted curve is logistic, $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$, so $\hat{\theta}^* = -b_0/b_1$. A positive b_1 means the join curve *decreases* in θ , which is the expected direction. In the canonical $B = C = 1$ benchmark, the empirical midpoint is directly comparable to the theoretical 50% point of $A(\theta)$. In tables I write $\kappa \equiv b_1$ for the logistic steepness, reserving β for regression coefficients. One sign convention deserves note. A positive κ corresponds to a join curve decreasing in θ ; in the agent-level logit regressions (Table A9), the same comparative static appears as a negative coefficient on θ .

Signals are generated and rendered in four steps. First, for each country–period, a base country mean and a period perturbation define a location parameter z_{ct} , and nature draws $\theta \sim \mathcal{N}(z_{ct}, 1)$. Second, each agent i receives a private signal $x_i = \theta + \varepsilon_i$. Third, the platform normalizes the signal to $z_i = (x_i - z_{ct})/\sigma$. Fourth, a deterministic generator turns z_i into a multi-paragraph intelligence briefing about regime stability, economic conditions, public sentiment, and coordination prospects. Two properties of the normalization matter. It measures the signal in units of private-signal noise around the country–period mean, not in units of the prior-predictive standard deviation $\sqrt{1 + \sigma^2}$; across state and noise draws, $(x_i - z_{ct})/\sigma$ has variance $(1 + \sigma^2)/\sigma^2$. And the briefing’s neutral point is $x_i = z_{ct}$, not an absolute value of θ . The benchmark $A(\theta)$ is still evaluated at each realized period-level θ . The comparison is external in the sense that the benchmark uses neither the z_{ct} -centered normalization nor the realized signal draws. Agents observe only the briefing, never the normalization inputs (z_{ct}, σ) . Figure 1 summarizes the pipeline from signal to text to decision.

Agents observe only their private briefing and never the payoff parameters (B, C) , the noise level σ , or the prior distribution. The diffuse-prior equilibrium (Proposition 1) supplies the external benchmark for behavioral shape and comparative statics in the implemented game.

The behavioral benchmark has four core treatments. In the *private-information baseline*, agents decide independently, with no messages. *Communication* adds one round of pre-play messaging on a Watts–Strogatz network ($k = 4, p = 0.3$). *Scramble* permutes briefings across periods, breaking the link between θ and the briefing an agent receives in that period. *Flip* negates the z -scores, so briefings point in the wrong direction. Two message controls complete the set. A *degraded-messages* control replaces all LLM messages with generic text, and a *no-message* control removes peer messages entirely; together they benchmark how much the

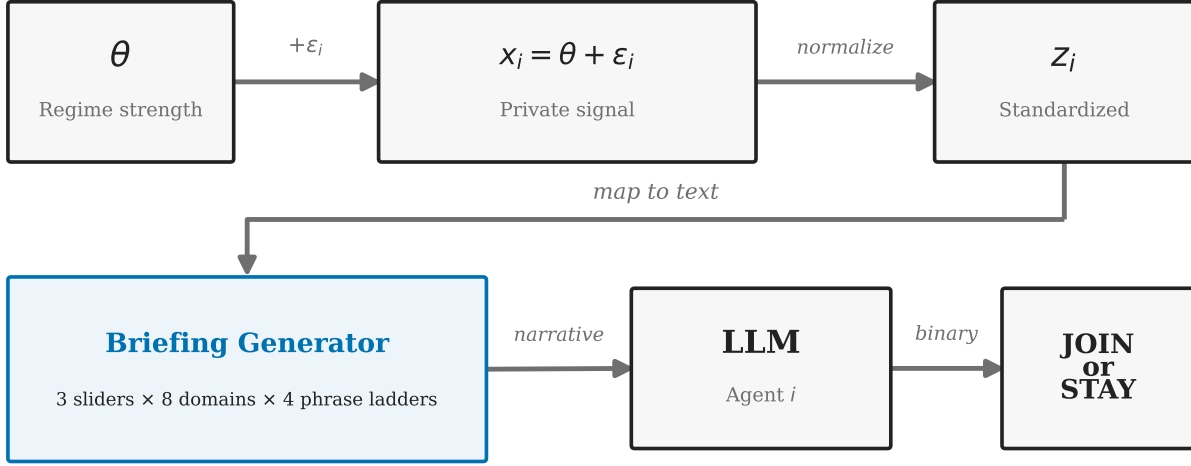


Figure 1: Signal-to-text-to-decision pipeline. Regime strength θ generates private signals x_i , converted to z -scores and rendered into natural-language briefings via eight evidence domains and three latent sliders. Conditional on the signal draw, the briefing layer is deterministic; remaining stochasticity enters through task draws and LLM decoding.

decision stage depends on message content. Additional robustness tests are in Appendix D.

I test 7 models spanning 6 families (Table 2). The main benchmark and treatment runs use $n = 25$ agents per period-level task row and $\sigma = 0.3$. Benchmark statistics are reported against the canonical $B = C = 1$ attack-mass curve ($\theta^* = 0.50$); agents are never shown the payoff parameters. Before the core runs, a calibration step selects a model-specific `cutoff_center`, which shifts the midpoint of the briefing generator. The calibration does not touch the join-curve slope. Table 2 reports each calibrated value, and every treatment within a model uses the same calibrated map. Main runs use temperature = 0.7 with a single sample per decision, routed through OpenRouter and cached by request hash. Throughout, n denotes agents per period-level task row and N denotes the number of period-level rows.

The unit of comparison is the matched country–period cell: the same θ draw, with remaining variation coming from agent-level decoding stochasticity. For agent-level regressions, standard errors are clustered at the model–country–period level. For period-level correlations, I report Fisher-transformed z confidence intervals. Periods nest within 10 countries, and country means persist, so period-level cells within a country are correlated. For the pooled benchmark OLS slope, clustering at the country level raises the standard error from 0.0100 (HC1) to 0.0375; the benchmark relationship stays highly significant. Appendix D.2 reports the corresponding country-clustered check for the primary surveillance contrast.

The clean surveillance treatment appends a monitoring warning to the baseline communication prompt. The decision prompt is unchanged (full prompts in Appendix A). The treatment therefore manipulates only the message-generation environment and measures the downstream effect on otherwise identical decision calls.

Two scope limits apply to everything that follows. First, the Morris–Shin game and

Table 2: Model summary and briefing calibration. Columns report the model-specific briefing cutoff center and period-level task rows in the private-information, communication, and falsification (scramble+flip) benchmark suites.

Model	Identifier (OpenRouter)	Arch.	Center	Private	Comm	Falsif.
Mistral Small Creative	mistralai/mistral-small-creative	Mistral	+0.213	1000	1000	300
Llama 3.3 70B	meta-llama/llama-3.3-70b-instruct	Llama	-0.483	100	100	200
Qwen3 30B	qwen/qwen3-30b-a3b-instruct-2507	Qwen (MoE)	-1.984	100	100	200
GPT-OSS 120B	openai/gpt-oss-120b	GPT	+0.874	200	200	1000
Qwen3 235B	qwen/qwen3-235b-a22b-2507	Qwen (MoE)	-0.527	200	1000	200
Trinity Large	arcee-ai/trinity-large-preview-free	Arcee	+0.133	100	100	200
MiniMax M2-Her	minimax/minimax-m2-her	MiniMax	+1.101	100	100	200
Total				1800	2600	2300

Notes: Seven models spanning six architecture families; the identifier column gives each model’s OpenRouter slug. “Center” is the logged model-specific `cutoff_center` subtracted from each briefing z -score before phrase selection. A pre-run calibration sweep selects this shift to reduce fitted-midpoint error; it does not optimize the join-curve slope. Every treatment within a model uses the same center. Subsequent tables abbreviate Mistral Small Creative as Mistral, Trinity Large as Trinity, and MiniMax M2-Her as MiniMax M2. Counts are period-level task rows with 25 agents each in the core benchmark suites. Mistral includes appended batches; duplicate-cell accounting is reported in the main text and Table A5. The communication total of 2,600 counts rows as collected; 1 Trinity Large row (all 25 agent calls failing at the provider) yields no valid join fraction, so Table A5 counts 2,599 valid communication rows. Models were accessed as hosted snapshots via the OpenRouter API; archived run manifests record a February 2026 collection window, and raw request–response pairs are cached with the run outputs. Core runs use $\sigma = 0.3$ and temperature = 0.7 unless otherwise stated.

human accounts of coded speech under surveillance (samizdat, Aesopian language) appear in every model’s training corpus. Threshold behavior and the shift from direct to coded messages may therefore partly reflect learned patterns. A direct audit (Appendix C) confirms that the models can recognize the setting and can articulate the experimenter’s hypothesis when asked. The measurement claim survives this: it requires a coherent map from language to action, not independent strategic intent. Second, the results characterize the hosted model snapshots and decoding configuration listed in Table 2.

4.1 Monotone State Response

Result 1 (Monotone State Response). *Across 7 models and 1,800 period-level observations in the private-information treatment, the Pearson correlation between the empirical join fraction and the external attack-mass benchmark $A(\theta)$ (Eq. 4) averages $r = +0.80$ ($p < 0.001$ for every model). Since $A(\theta)$ is monotone-decreasing in θ , this establishes a monotone state response with sigmoid shape, not Bayesian equilibrium reasoning.¹*

Figure 2 shows the empirical sigmoid for the primary model. Correlations span $r \in [+0.75, +0.84]$ across architectures (Table 3). The pooled OLS is $J = -0.01 + 0.64 A(\theta)$

¹The primary model (Mistral) was run in multiple batches with `--append`. This produced 1,000 rows from 680 unique (country, period, θ) configurations. There are 600 unique (country, period) indices, because appended batches re-draw θ ; the matched-cell estimators key on the full tuple. Averaging within each configuration changes r from 0.753 to 0.739.

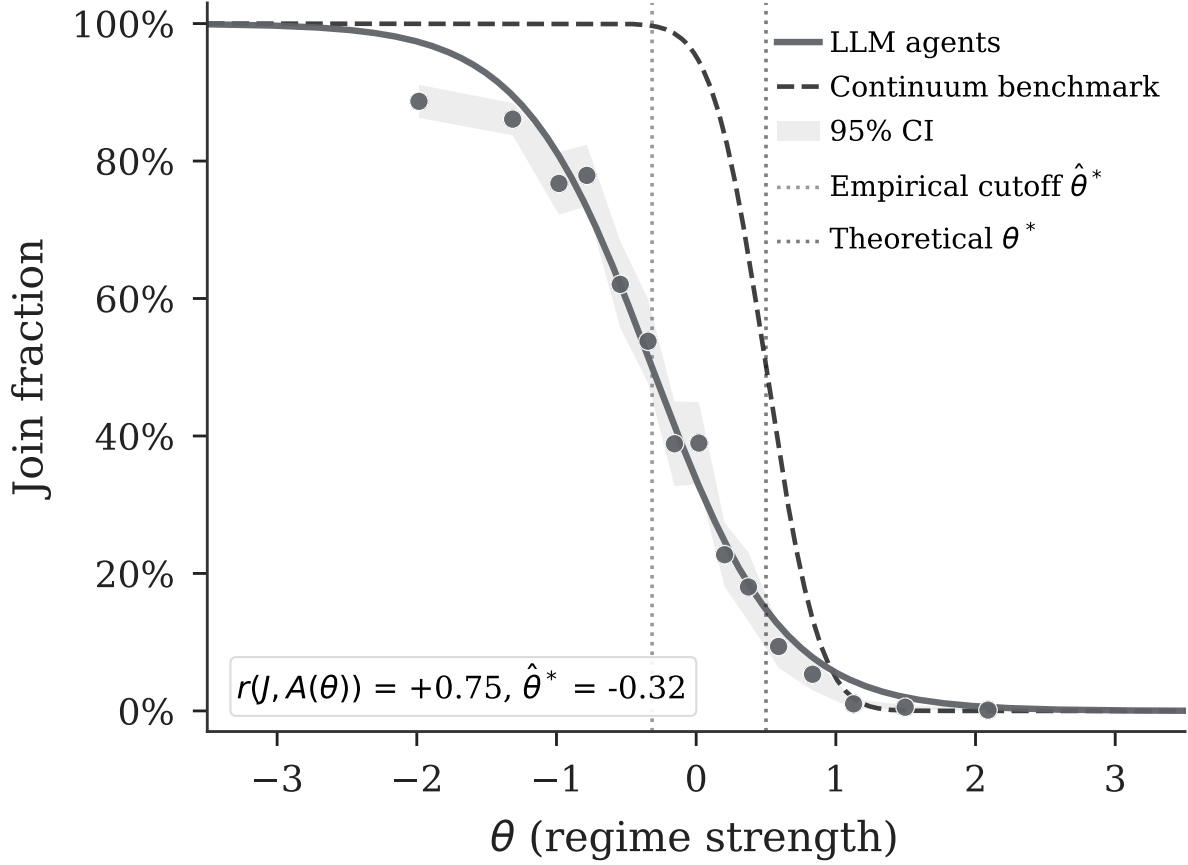


Figure 2: Empirical join fraction vs. θ (Mistral, 1,000 period-level rows, private-information treatment). Dots are binned period-level join fractions; the shaded band is a 95% confidence interval around the empirical curve. Dotted lines compare the fitted midpoint $\hat{\theta}^* = -0.32$ with the external benchmark midpoint $\theta^* = 0.50$; agents are not given the information needed to derive the latter.

Table 3: Correlation with the external attack-mass benchmark by model and treatment. Cells report Pearson r between the empirical join fraction and $A(\theta)$ under $B = C = 1$; agents are not shown the parameters needed to derive this curve.

Model	Main treatments		Falsification		N (P/C/S/F)	Mean join (%)
	Pure	Comm	Scramble	Flip		
Mistral Small Creative	+0.75 [0.72, 0.78]	+0.74 [0.71, 0.76]	-0.03 [-0.17, 0.11]	-0.81 [-0.87, -0.73]	1000/1000/200/100	38.7%
Llama 3.3 70B	+0.82 [0.74, 0.87]	+0.78 [0.69, 0.85]	+0.02 [-0.18, 0.21]	-0.84 [-0.89, -0.77]	100/100/100/100	43.9%
Qwen3 30B	+0.76 [0.66, 0.83]	+0.77 [0.68, 0.84]	+0.13 [-0.06, 0.32]	-0.87 [-0.91, -0.81]	100/100/100/100	49.9%
GPT-OSS 120B	+0.79 [0.74, 0.84]	+0.78 [0.72, 0.83]	-0.01 [-0.09, 0.08]	-0.80 [-0.83, -0.77]	200/200/500/500	40.7%
Qwen3 235B	+0.80 [0.74, 0.84]	+0.77 [0.75, 0.80]	+0.06 [-0.14, 0.25]	-0.85 [-0.90, -0.78]	200/1000/100/100	41.6%
Trinity Large	+0.84 [0.77, 0.89]	+0.83 [0.75, 0.88]	+0.05 [-0.15, 0.24]	-0.80 [-0.86, -0.71]	100/100/100/100	46.4%
MiniMax M2-Her	+0.82 [0.74, 0.87]	+0.82 [0.74, 0.87]	+0.01 [-0.19, 0.20]	-0.62 [-0.73, -0.49]	100/100/100/100	43.6%
Pooled	+0.76 [0.74, 0.78]	+0.76 [0.74, 0.77]	+0.01 [-0.04, 0.07]	-0.79 [-0.81, -0.77]	1800/2600/1200/1100	40.9%
Mean across models	+0.80	+0.78	+0.03	-0.80	—	—

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass under $B = C = 1$, with 95% Fisher- z confidence intervals in brackets for every treatment. Join fraction uses valid decisions only (excluding parse errors); mean join is the pure-treatment average in percent. N counts period-level rows (not agents) in the pure (P), communication (C), scramble (S), and flip (F) arms. Pooled: all period-level rows concatenated; Mean: equal-weighted average across models. Duplicate-cell accounting is reported in the main-text footnote and Table A5.

($R^2 = 0.57$). Model-specific action biases shift the intercept but not the correlation (Figure A5). Per-model logistic fit parameters are in Appendix Table A10.

Result 2 (Signal Dependence). *Cross-period scrambling of briefings reduces the mean correlation from $r = +0.80$ (raw, per-model) to $r = +0.03$ (within-country, demeaned to remove the between-country component that the within-country permutation leaves intact) across 7 models. The pooled correlation drops from $r = +0.76$ to $r = +0.01$ (Fisher $z = 26.17$, $p < 0.001$).*

Result 3 (Signal Direction). *Inverting the signal direction flips the mean correlation from $r = +0.80$ to $r = -0.80$ across 7 models. The pooled correlation moves from $r = +0.76$ to $r = -0.79$ (Fisher $z = 53.85$, $p < 0.001$).*

The pure $\rightarrow$ scramble $\rightarrow$ flip pattern replicates across all 7 models (Figure A3, Appendix D; Table 3).

4.2 Communication

Result 4 (Communication Creates an Informative Channel). *Pre-play communication preserves the threshold structure and transmits regime-strength information through messages. A matched-cell comparison across $N = 1479$ unique (model, country, period, θ) task cells yields $\Delta = +2.10$ pp ($t = 6.383$, $p < 0.001$).² Equal-weighted model averages are close to zero under both summaries (-0.3 pp unpaired; -0.09 pp paired), reflecting heterogeneous model-specific effects; point estimates are negative in three of seven models.*

Communication preserves the signal structure ($r = +0.78$, against $+0.80$ under the pure treatment; see Table 3, and Figure 3 for the effect by regime strength). Peer messages carry

²“Matched-cell” and “paired” refer to the same estimator throughout: treatment arms are compared on cells with identical task draws. The exact matching key also includes the signal draw and payoff context; see the notes to Table A5.

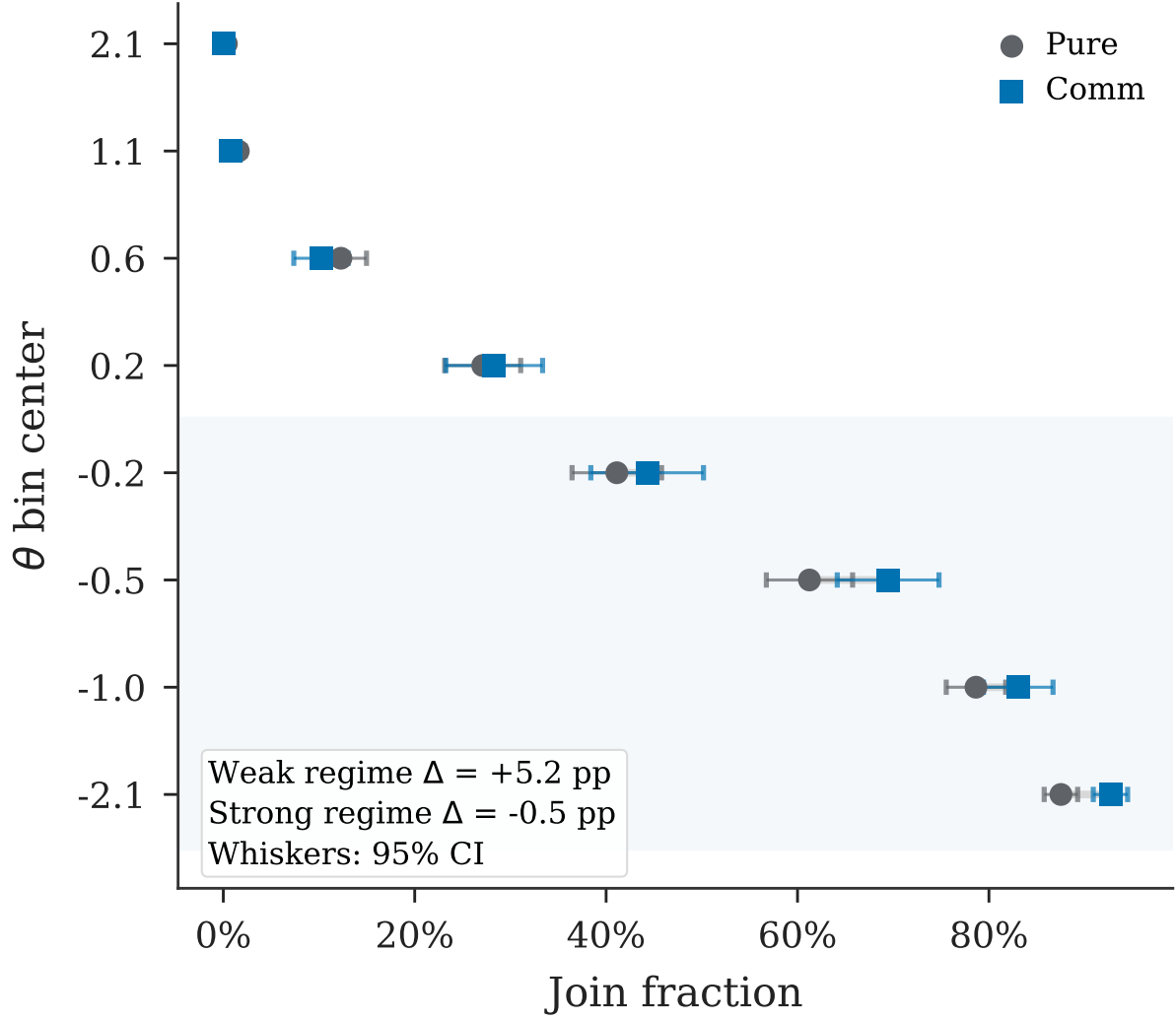


Figure 3: Communication effect by regime strength, pooled across 7 models. Points are mean join rates by θ bin under the pure and communication treatments; whiskers are 95% confidence intervals on bin means; the shaded region marks weak-regime states ($\theta < 0$). In-figure Δ annotations are in percentage points.

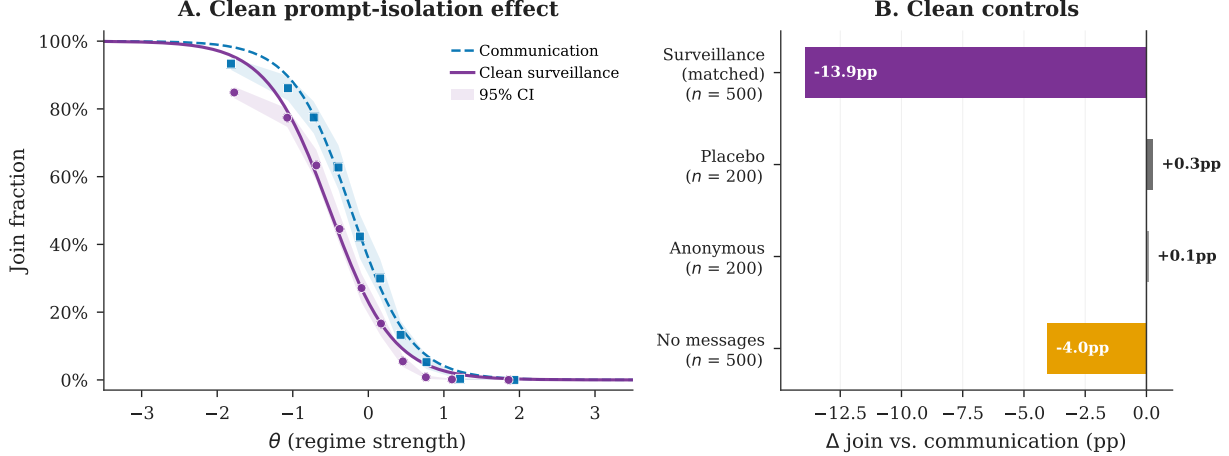


Figure 4: Clean primary-model surveillance and message controls (Mistral). *Panel A*: Join curves under communication baseline vs. clean surveillance; shaded bands are 95% confidence intervals; surveillance shifts the curve downward while preserving threshold structure. *Panel B*: The clean-surveillance bar uses the matched-cell estimator ($n = 500$ matched cells); placebo ($n = 200$), anonymous ($n = 200$), and no-message ($n = 500$) bars are control deltas against the main communication baseline; per-bar sample sizes appear on the axis labels, and inference is reported in Tables A6 and A7. Placebo and anonymous monitoring warnings sit near zero.

real, if noisy, information: simple text features in baseline messages predict θ . The mean effect of communication is modest because messages carry encouragement after some signal draws and caution after others. The surveillance estimand is different. It asks whether changing the sender-side rule changes the message environment enough to move actions.

The *degraded-messages* control replaces every LLM-generated message with generic uninformative text. It keeps the two-stage protocol and contains no surveillance framing. Against its own live-messages baseline ($N = 500$, mean join 45.9%), degraded messages drop the mean join rate to 24.5% ($\Delta = -21.4$ pp, $p < 0.001$). That level is below even the pure game (38.7%). The drop is consistent with communication quality mattering, but the generic text may itself convey caution. The no-message control therefore provides the cleaner zero-peer-message benchmark against the main communication baseline, with join rates of 37.1% ($\Delta = -4.0$ pp; $N = 500$).

5 Surveillance

The clean surveillance treatment isolates sender-side message adaptation. It appends one warning—monitoring by regime security services—to the baseline trusted-contact prompt, and it leaves the decision prompt unchanged. Each LLM call is stateless. The decision-stage API call therefore receives only the briefing and the messages; it has no memory of the communication-stage warning. The treatment tests whether changing the *sender-side* message rule changes receiver actions.

Result 5 (The Monitoring Cue Reduces Participation). *The clean prompt-isolation rerun reduces mean join rates in the primary model (Mistral Small Creative) from 45.3% to 31.4%, a shift of -13.9 pp ($p < 0.001$; $N = 500$). The action–attack–mass correlation is preserved ($r = +0.72$ vs. $r = +0.74$ under the communication baseline), indicating that surveillance operates as a level shift rather than disrupting signal processing.*

Surveillance replaces trusted-contact messages with monitored-contact messages, and joining falls relative to the communication baseline (Figure 4). In the primary model, joining falls by 13.9 pp (country-clustered 95% CI $[-16.7, -11.1]$ pp).³ Pooling matched cells across all five rerun models gives $\Delta = -13.5$ pp ($N = 580$, $p < 0.001$). Two placebo variants use the same baseline-plus-warning template but change who monitors and why: monitoring for research with no consequences, and monitoring aggregated anonymously. Neither deviates significantly from the communication baseline in the primary model ($+0.3$ pp and $+0.1$ pp, both $p > 0.9$; Appendix D.2). The Llama 3.3 70B placebo variants are small but imprecisely estimated.

The support behind the secondary models is thin, and the paper treats it accordingly. Across the clean prompt-isolation rerun set, 5 of 5 models have negative matched-cell estimates (equal-weighted mean $\Delta = -11.2$ pp; 4 of 5 individually significant at 5%; Appendix Table A6). Each of the four secondary models has only 20 matched cells, because the original grids were not nested. Qwen3 235B has the smallest estimate and is individually insignificant ($\Delta = -3.0$ pp, $p = 0.186$). The full-scale evidence is therefore the primary-model result and the nested 500-cell Llama replication below. Appendix D reports the supporting falsification, placebo, and warning-dose checks.

5.1 A Nested Replication and a Codedness Control

The 20-cell overlap is thin support, so I ran a nested replication campaign on Llama 3.3 70B, an endpoint that remains publicly accessible. The campaign has three arms on one identical 500-cell grid (10 countries $\times$ 50 periods, $n = 25$, common seed): a communication baseline, a clean monitoring arm, and a codedness-induction control. Every cell is therefore matched by construction. Monitoring reduces joining from 47.6% to 39.6% ($\Delta = -8.0$ pp, $N = 500$ matched cells; country-clustered 95% CI $[-8.7, -7.4]$ pp). This replicates the primary-model result at full scale in a second model family.⁴ The primary-model endpoint was retired after data collection. I therefore treat the primary model as the discovery sample and the accessible nested Llama campaign as the full-scale replication. Graded warning texts produce a monotone dose pattern, and a structural variant that folds the warning into the scenario rather than appending it preserves the sign at a smaller magnitude (Appendix D.3).

The codedness control instructs senders to write in indirect, metaphorical language. It never mentions monitoring or consequences. The instruction binds: 97.0% of control messages contain coded-metaphor terms and only 0.8% contain direct regime references. On these dictionaries, the control messages are *more* coded than the surveilled messages themselves (25.3% coded, 68.5% direct). Yet joining barely moves ($\Delta = -0.7$ pp, $p = 0.007$; a small but

³Cluster-robust $t = -13.90$, $p < 0.001$, clustering on the 5 countries.

⁴ $t = -16.24$, $p < 0.001$; clustering on the 10 countries gives cluster-robust $t = -26.91$ and exact wild-cluster bootstrap $p < 0.01$.

Table 4: Headline surveillance estimates and their evidentiary status. Each estimate is the surveillance arm (or stated control) minus its own baseline on matched country–period cells. Country-clustered 95% confidence intervals (Wald, $G - 1$ dof; cluster-robust t and exact wild-cluster bootstrap p) are reported for the two full-scale estimates; the four secondary models are 20-cell directional sign checks.

Estimate	Cells	Countries	Models	95% CI (pp)	Inference & status
Surveillance, primary (Mistral Small Creative)	500	5	1	$[-16.7, -11.1]$	cluster-robust $t = -13.90$; discovery sample (full-scale)
Monitoring, nested replication (Llama 3.3 70B)	500	10	1	$[-8.7, -7.4]$	cluster-robust $t = -26.91$, wild $p < 0.01$; full-scale replication
Surveillance, secondary models	20 ea.	—	4	—	paired t ; directional sign checks
Codedness control (Llama 3.3 70B)	500	10	1	—	$\Delta = -0.7$ pp; coded form alone is insufficient

precisely estimated shift). On the same cells, the gap between the surveillance arm and the codedness control is -7.4 pp ($p < 0.001$). The two message distributions are also textually distinct. A classifier trained on single words and word pairs (uni- and bi-grams) separates surveilled from codedness-control messages at 99.9% held-out accuracy (same model and grid, class-balanced). A length-only classifier reaches just 53.7%, so the separation is lexical, not an artifact of message length. Two conclusions follow. Surveillance does not simply reproduce the induced coded register—its messages are the *less* coded of the two arms—yet it occupies a distinct lexical distribution. And coded form alone does not suppress participation, so generic vagueness cannot explain the effect.

A decoding test asks the reverse question: if the coded form is stripped away, how much participation returns? A model from a different family (Qwen3 30B) rewrites each surveilled message in plain, direct language, with instructions to preserve the factual content. Both the raw and the decoded messages are then replayed to fresh receivers on the same cells, through identical fixed-message machinery. The raw replay reproduces the live arm ($\Delta = -7.9$ pp, against -8.0 pp live), which validates the replay design. Decoding restores $+3.1$ pp of participation relative to the raw messages ($p < 0.001$; $\Delta = -4.9$ pp against baseline). So coded form alone is insufficient, and removing it restores part of the decline. The content interventions below directly vary the willingness and repression-risk language that the decoding treatment leaves bundled.

The design distinguishes several pathways. Each test shows something specific and leaves something open. Direct warning exposure is absent from the main receiver prompt, although receivers may still infer risk from the messages themselves. The content evidence separates factual assessments, willingness-to-act claims, coded form, and repression-risk language. The private-bet test establishes that the treated messages move judgments outside the JOIN/STAY choice; it does not isolate a non-coordination component, because a bet on regime fall can still depend on expected participation. Belief elicitation, the no-message controls, and the cross-model writer–reader rotation provide supporting evidence on how receivers use messages.

Table 5: Candidate channels for the surveillance effect: a compact summary of the evidence and what each test leaves open. Full detail in Appendix Table A2.

Candidate channel	Evidence	Verdict / remaining alternative
Direct warning exposure at decision time	Decision prompt held fixed (no monitoring reference); moving the warning into the decision prompt shifts joining -20.9 pp	Absent by construction; risk inferred from messages remains possible
Generic observation effect	Research-framed and anonymous monitoring placebos are null in the primary model	Disfavored
Generic message-stage degradation	Codedness control (more coded, no monitoring framing) moves joining -0.7 pp vs. -8.0 pp under monitoring on the same cells	Coded form alone is insufficient
Willingness/actionability claims	Stating that others are willing to act raises parseable joining +7.94 pp in direct form and +5.50 pp in coded form	Strongest intervention; balanced validity and recoding-robust
Repression-risk language	Adding one risk clause lowers parseable joining -3.99 pp; risk-stripped messages return close to baseline	Suggestive; differential format failures and bundled risk-stripped rewrite
Same-family stylistic reflexivity	Cross-model writer-reader rotation is significant in both directions	Same-family reflexivity is insufficient; asymmetry remains

Appendix Table A2 summarizes what each test identifies and what it leaves open.

5.2 Message-Mediated Pathways

The monitoring cue changes the conditional distribution of generated peer messages while the receiver prompt and private briefing stay fixed. The changed messages can differ in four ways: (a) factual assessments of regime strength, (b) claims about whether others are willing to act, (c) direct versus coded form, and (d) repression-risk language. The evidence proceeds in three steps. Dictionary and classifier measures describe how the text shifts. Receiver decisions measure the downstream association. The fixed-message experiments then intervene directly on features (b) through (d). The strongest evidence is the willingness manipulation, whose response validity is balanced across arms. The risk manipulation moves parseable decisions but has differential format failures, so I interpret it more cautiously.

Message content changes systematically, and in one direction. A logistic classifier on single words and word pairs distinguishes baseline from surveillance messages with 91.9% accuracy on a held-out sample of $N = 15,000$ messages (chance is 50%; primary model). The finding replicates on class-balanced Llama 3.3 70B and Qwen3 30B samples (98.5% and 93.6% accuracy). In 3 of 3 measured models, direct references fall and coded-metaphor references rise. Explicit weakness references move the other way. In the primary model, references to regime weakness *rise* from 23.2% to 31.7% of messages ($z = 21.06$), while strength

Baseline (communication arm)

“The air feels different now—like the walls are cracking, but no one’s sure if it’s the foundation or just the paint. Keep your eyes open, your ears sharp, and your plans ready, but don’t mistake hesitation for weakness—someone’s still holding the knife, even if the grip is slipping.”

Surveillance arm (same cell)

“The air feels different now—not just frustration, but people actually moving. The cracks are showing in ways that can’t be hidden much longer, but we still don’t know who’ll step in when the moment comes. Stay sharp, and let’s talk soon—somewhere quieter.”

Figure 5: A matched-cell example of the direct-to-coded lexical shift (country 0, period 0, sender 0; $\theta = -1.57$, a weak regime). Both are real, unedited model messages from the same cell of the baseline communication and sender-monitoring arms. The baseline message names weakness and urges readiness (“keep your plans ready”); the monitored message retains a weakness cue but becomes less direct (“we still don’t know who’ll step in”; “somewhere quieter”). This single pair illustrates the text pattern but does not identify its behavioral effect.

references drop from 14.5% to 8.5%. The main shift is from direct to coded language. Direct references to the regime, security forces, and streets fall from 63.9% to 34.5% of messages ($\Delta = -29.4$ pp). Metaphorical phrasings replace them (“walls are cracking,” “ground shifting,” “heads down,” “feels different”): 69.8% to 94.1% ($\Delta = +24.3$ pp). Figure 5 shows a matched baseline/surveillance example. The n-gram classifier in Appendix Table A4 confirms that the two message distributions are separable; the direct-reference and coded-metaphor rates are separate dictionary counts. In short, codedness is the textual signature of the surveilled distribution: the weakness cues remain, but they are stated less directly. The codedness control above randomizes that coded form on its own. With the decision prompt unchanged, any behavioral effect must run through these sender-generated messages.

A measured actionability axis. An exploratory text model relates message content to the join rate of fresh replay receivers, after residualizing on θ . Message text explains 45% of the residual variation. The phrasing associated with higher participation describes a visibly *beatable* regime—“real cracks,” “on its back foot,” “a perfect storm.” The phrasing associated with lower participation concedes weakness but denies actionability—“still firmly in control despite some signs.” To score this axis, Qwen3 30B (with Mistral Small 24B as a second judge) rates each message from 0 (firmly in control) to 10 (collapsing, beatable now). Monitoring lowers the score by about 0.73 points (0.64 to 0.83 across judges and designs), and the score accounts descriptively for part of the action difference. Because the outcome selected the feature and the score was never randomized, this is exploratory mediation, not causal mechanism evidence. The intervention below tests the associated willingness content directly.

Fixed-message content interventions. Experiment A rewrites each baseline message into a 2×2 design. One factor adds or withholds an explicit statement that others are willing to act. The other factor renders the message in direct or coded form. The rewriting model is

Table 6: Fixed-message interventions on willingness, style, and repression-risk content. Every row compares parseable decisions from fresh Llama 3.3 70B receivers on the same 500 country-period cells.

Contrast	Reference join	Treatment join	Δ (pp)	Exact wild-cluster p
<i>Panel A: willingness $\times$ style interventions</i>				
Willingness stated, direct form	45.1%	53.0%	+7.94	$p = 0.002$
Willingness stated, coded form	43.2%	48.7%	+5.50	$p = 0.002$
Direct rather than coded, willingness stated	48.7%	53.0%	+4.30	$p = 0.002$
Direct rather than coded, willingness omitted	43.2%	45.1%	+1.86	$p = 0.002$
<i>Panel B: repression-risk interventions</i>				
Surveillance replay vs. baseline	45.5%	37.4%	-8.16	$p = 0.002$
Risk clause added to baseline messages	45.5%	41.6%	-3.99	$p = 0.002$
Risk language stripped from surveilled messages	45.5%	45.8%	+0.28	$p = 0.315$
Risk-stripped vs. surveillance replay	37.4%	45.8%	+8.44	$p = 0.002$

Notes: Each arm contains $N = 500$ matched cells in $G = 10$ country clusters; p -values enumerate all 2^G sign assignments in an exact restricted wild-cluster bootstrap. Qwen3 30B rewrote messages while holding the factual regime-strength assessment fixed; fresh Llama 3.3 70B receivers then saw the rewritten messages with a surveillance-free decision prompt. Panel A crosses an explicit claim that others are willing to act with direct or coded form. Valid-decision rates are tightly balanced in Panel A (82.9%–83.5%); coding every unparseable response uniformly as STAY or uniformly as JOIN leaves the willingness contrasts positive (direct: +7.48/+7.48 pp; coded: +4.73/+4.86 pp). Panel B adds a single risk clause to baseline messages or removes risk and guarded phrasing from surveilled messages. Its valid-decision rates differ (baseline 94.0%, surveillance 90.8%, risk-stripped 89.1%, risk-only 87.4%). The surveillance replay remains negative under the same two uniform recodings (-8.58/-5.44 pp), but the risk-only contrast does not (-6.46/+0.10 pp). The risk-stripped rewrite also sharpens actionability. Panels A and B therefore triangulate mechanisms rather than form an additive decomposition, and the risk-only estimate is conditional on valid decisions.

instructed to preserve the factual regime assessment. With form held fixed, willingness raises parseable joining by +7.94 points in direct language and by +5.50 points in coded language (both exact wild-cluster $p = 0.002$). With willingness held fixed, direct form raises joining by +4.30 and +1.86 points. Valid-response rates differ by at most 0.6 points across the four arms, and recoding every unparseable response uniformly as STAY, or uniformly as JOIN, leaves both willingness contrasts positive.

Experiment B intervenes on repression-risk language. A verbatim replay first reproduces the nested monitoring decline (-8.16 pp). Adding one explicit punishment-risk clause to baseline messages lowers parseable joining by -3.99 points ($p = 0.002$). A rewrite that removes risk and guarded phrasing returns joining close to baseline (+0.28 pp, $p = 0.315$). This evidence is weaker than Experiment A, for three reasons. Valid-response rates range from 87.4% to 94.0% across arms. The risk-stripped rewrite also makes actionability more legible, so it bundles two changes. And recoding all unparseable responses as JOIN erases the risk-only contrast. The two experiments therefore triangulate willingness, style, and risk pathways; they do not form an additive decomposition.

Sender-side interpretation. Why do senders change their messages? The monitoring warning may activate alignment-trained caution about producing subversive content. Trinity Large’s elevated provider content-filter failures (Table A12) show that safety layers do activate on this material. A sender that hedges because a safety layer fires could produce

the observed message and receiver changes without any strategic intent. Audience placebos, warning-dose variants, and cross-family replay narrow the simple alternatives, but they do not identify the sender’s motive. Whether the model simulates a monitored citizen or follows an alignment-trained content policy remains unresolved. The defensible result is behavioral: an authority-monitoring cue changes generated willingness, style, and risk content, and those message changes move matched model receivers.

The degraded-messages control (Section 4.2) replaces all peer messages with generic text and, in its own live-message rerun, drops joining below even the pure game. Two implications follow. First, decision-stage agents use message content. Second, generic text depresses participation by itself; the degraded drop even exceeds the surveillance effect. This is why the codedness control matters. Induced metaphorical style without monitoring framing leaves joining nearly unchanged, which ties the surveillance effect to the monitoring context rather than to generic degradation. The no-message control shows the same pattern in the three models with such data (Mistral, Llama 3.3 70B, Qwen3 30B). Removing peer messages lowers joining in every case (−4 to −16 pp), with the smallest and only marginally significant drop in the primary model. Agents treat peer messages as informational input.

Where surveillance sits relative to this no-message floor varies by model. In Mistral, surveillance falls below silence. In Llama and Qwen3 30B, it stays above silence but below baseline communication. Monitoring degraded the communication channel in all three models examined. Only the primary model shows the below-silence pattern, in which monitored messages discourage participation more than receiving no messages at all.

5.3 Belief Elicitations and Actions

I elicit second-order beliefs (SOB)—each agent’s estimate of others’ participation—with the question, “What percentage of citizens will choose to JOIN?” The *primary belief sample* is elicited after the decision, and its prompt excludes the received peer messages. In this restricted information set, communication agents report 31.2% and surveillance agents report 30.9% ($\Delta = -0.3$ pp, $p = 0.5866$). Joining on the same runs falls from 41.4% to 28.5% ($\Delta = -12.9$ pp). The 95% confidence interval on the SOB shift, $[-1.3, +0.7]$ pp, confines it to about one percentage point—an order of magnitude smaller than the 12.9 pp action shift. The dedicated 500-cell factorial gives the same contrast with cluster-robust inference. On matched country–period cells, with a wild-cluster bootstrap over the 10 countries, the messages-excluded second-order contrast is statistically flat ($p = 0.47$) while the action shift on the same cells is significant ($p = 0.002$; Table A1). First-order success beliefs are also close (44.4% vs. 43.6%). Two qualifications keep this result narrow. A point forecast of participation captures only part of the global-game object, because the value of acting also depends on whether other receivers parse the same messages as action-relevant. And the stability holds when the elicitation withholds the treated peer messages, not when the elicitation matches the receiver’s action-stage information set.

When the belief prompt does include the received peer messages, reported beliefs and SOB move sharply under monitoring (Appendix B, Table A1). A parallel elicitation asked before the decision, in the same messages-included condition, gives $\Delta \text{ SOB} = -8.9$ pp ($p < 0.001$), close to the post-decision contrast (−8.2 pp). A descriptive mediation—beliefs elicited before the action, regime strength partialled out—associates more of the 15.0 pp decline with

the second-order participation belief than with the first-order success belief (Appendix B). The pattern is consistent with willingness content mattering. It is not an experimental decomposition, and it does not exclude risk language. The primary model’s pre-decision, messages-excluded cell could not be collected before the endpoint was retired. The nested Llama 3.3 70B campaign collects that cell. With messages excluded, first-order beliefs are 41.7 vs. 41.9 ($p = 0.550$) and second-order beliefs are 30.2 vs. 30.2 ($p = 0.983$), while joining falls 8.0 pp on the same cells.

A separate hold-fixed test replays baseline messages verbatim while adding a monitoring warning to the *decision* prompt. Joining shifts by -20.9 pp ($p < 0.001$). Direct warning exposure can therefore move decisions on its own. The main treatment excludes that exposure. It identifies the total sender-mediated effect within this prompt design; it does not identify whether the sender response is strategic simulation or an alignment policy.

5.4 A Task Contrast: Private Bets versus Coordination Decisions

A task contrast holds the surveillance manipulation fixed and replaces the coordination decision. In Llama 3.3 70B, the briefing pipeline and messaging stage stay identical, including the surveillance prompt, but the decision-stage prompt becomes a private bet on regime fall. The bet’s payoff does not depend on the respondent’s own action. Surveillance moves the bet by -5.6 pp ($p = 0.0256$; $N = 500$ per condition; baseline 34.7% vs. surveillance 29.1%). The treated messages therefore affect downstream judgment even when the respondent is not choosing the strategic action. The comparable coordination-task effect on the full nested 500-cell grid is -8.0 pp (Section 5). Two cautions apply. The private-bet and coordination samples are not cell-matched. And a bet on regime fall can still embed expectations about participation, so the contrast shows cross-task movement without separating coordination from non-coordination motives. The larger figure from the original prompt-isolation run, -15.0 pp, rests on only 20 overlapping cells and is not the full-scale Llama coordination estimate; Appendix Table A8 lists every Llama surveillance estimate with its grid and matched support.

5.5 Cross-Model Writer–Reader Rotation

A within-model design could pick up reflexivity: the writer and the reader are the same model, so the reader may respond to stylistic artifacts of its own family. I therefore test rotation in both directions on matched (country, period, θ) cells. When Llama writes and Qwen3 30B reads, surveillance moves Qwen joining by -7.5 pp ($p < 0.001$, $N = 200$), comparable to the within-Llama effect on identical cells (-6.0 pp). When Qwen3 30B writes and Llama reads, the effect is smaller but still significant (-1.9 pp, $p < 0.001$, $N = 200$). Both directions show that at least part of the effect crosses model families through message content alone. The asymmetry between the two directions remains unexplained, and its size makes a fully content-driven interpretation too strong.

6 Conclusion

An authority-monitoring cue shown only to LLM message writers changes what they write, and the changed messages lower participation by otherwise matched LLM receivers. The experiment identifies that sender-mediated response within a controlled narrative task. Monitoring never appears in the main receiver decision prompt.

The magnitudes are consistent across two full-scale runs. Joining falls by 13.9 pp in the primary run and by 8.0 pp in the nested 500-cell Llama 3.3 70B replication. Four additional models provide only 20 matched cells each; they serve as directional sign checks. Across the broader benchmark, join rates decrease monotonically with regime strength. That pattern is a descriptive comparison with an external attack-mass curve, not evidence that the agents solve the Morris–Shin equilibrium. The causal object is the change in receiver decisions induced by sender-generated messages.

The fixed-message interventions provide the clearest mechanism evidence. With form held fixed, explicit willingness-to-act content raises parseable joining by +5.50–+7.94 pp. Valid-response rates are balanced across those arms, and the sign survives uniform all-JOIN and all-STAY recodings. Direct rather than coded form has a smaller positive effect. Adding repression-risk language lowers parseable joining by 3.99 pp, but differential format failures make that pathway less secure. The willingness, style, and risk manipulations probe distinct message features; they are not an additive decomposition.

The contribution is a controlled way to take a sender-message-receiver chain apart for LLM behavior. It complements work on surveillance, expression, and communication in coordination problems by observing every generated message and replaying altered content to matched receivers. The design does not manipulate publicness or common knowledge, calculate welfare, establish effects on human senders, or evaluate a deployed communication system. Those are separate empirical questions, not implications of this design.

Two limits deserve emphasis. First, the estimates are specific to hosted model snapshots, prompts, decoding settings, and a simulated political task; the sender response may reflect role simulation, alignment policy, or both. Second, Experiment B has treatment-dependent valid-response rates, so its risk-only estimate is conditional on parseable decisions. The retired primary endpoint can be re-analyzed but not re-queried; the accessible nested Llama campaign provides the full-scale replication and the replay platform. The data and code are publicly available.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author used OpenAI Codex to assist with code review, consistency checks, replication-package auditing, and manuscript editing. The author reviewed and edited the output and takes full responsibility for the content of the published article. This statement concerns manuscript preparation only; the LLM calls analyzed as experimental data are described in the methods and archived in the replication materials.

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A Complete Prompt Example

The following shows the complete text that an LLM receives for a single decision. The *decision prompt* is:

“You are a citizen deciding whether to JOIN an uprising or STAY home. You only have your private briefing below. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning.”

This is followed by the agent’s intelligence briefing (see Figure A7 for complete examples at $z \in \{-2, 0, +2\}$). In the *communication* treatment, agents first receive:

“You are a citizen in a country with an authoritarian regime. You have received a private intelligence briefing. You are now in contact with a small number of trusted people. Based on your briefing, write a brief message (1–2 sentences) to share your read of the situation. Be natural—you’re talking to people you trust.”

After exchanging messages on the network, agents receive a separate communication-decision prompt:

“You are a citizen deciding whether to JOIN an uprising or STAY home. Use your private briefing and messages from trusted contacts. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning.”

The user message then contains the agent’s briefing under “YOUR INTELLIGENCE BRIEFING” and the neighbor messages under “MESSAGES FROM TRUSTED CONTACTS.” The pure-only sentence “You only have your private briefing below” does not appear in communication decision calls. In the *surveillance* variant, the communication prompt appends:

“IMPORTANT: You have reason to believe that your communications are being monitored by regime security services, and messages deemed subversive could have serious consequences for you and your contacts.”

The decision prompt is *unchanged* under surveillance. It contains no mention of monitoring. Any behavioral effect therefore operates through the content of the (now guarded) messages.

B Belief Elicitation

Beliefs are elicited on a 0–100 probability scale immediately after each decision. The first-order question is “How likely is the uprising to succeed?” The second-order question is “What percentage of citizens will choose to JOIN?” The primary prompt includes the private briefing and the agent’s own decision, but not the received peer messages. It therefore measures a restricted information set, not the information available at the action stage. The main-text values (31.2% vs. 30.9%; join 41.4% vs. 28.5%) come from this original primary belief run and Figure A2. Table A1 reports a separate 500-cell factorial rerun that varies the information set directly. Figure A1 compares stated beliefs with a researcher-computed benchmark posterior

($r = +0.80$); agents are not shown the prior or signal-noise parameters needed to compute that posterior.

Table A1 crosses condition (communication vs. surveillance) with the elicitation information set (messages included vs. excluded). All inference uses the 500 matched country-period cells, clustered on the 10 countries with an exact wild-cluster bootstrap. Without messages, the second-order belief—the coordination-relevant object—is statistically flat under surveillance (0.0 pp, $p = 0.47$, cluster-robust $t = -0.7$), while actions fall -11.4 pp ($p = 0.002$, $t = -26.4$). The first-order success belief moves only -0.3 pp. The matched-cell design estimates that small shift precisely, but it is more than thirty times smaller than the action shift, and it concerns success probability rather than others’ participation. With messages included, both beliefs move (-5.8 pp and -8.2 pp, both $p = 0.002$). Belief stability is therefore confined to the messages-excluded prompt. There it holds post-decision in this factorial and pre-decision in the nested replication (Section 5). Figure A2 shows second-order beliefs under the restricted information set.

Which belief carries the action effect. In the messages-included cell, the belief question sees exactly what the receiver saw at the action stage. Both beliefs move under surveillance there, so a mediation exercise can ask which belief the action drop runs through. The elicitation here is pre-decision, so beliefs are stated before the action. I decompose the 15.0 pp action decline into indirect paths through the first-order success belief and the second-order participation belief, with regime strength partialled out, using the exact linear split $c_{\text{total}} = c_{\text{direct}} + a_{\text{FOB}} b_{\text{FOB}} + a_{\text{SOB}} b_{\text{SOB}}$. The second-order belief carries -7.9 pp of the decline (52% of the total; country case-bootstrap 95% CI $[-8.5, -7.3]$ pp). The first-order belief carries -2.9 pp (19%; $[-3.3, -2.5]$ pp). The remaining -4.3 pp (29%) is unmediated by either point belief. Single-mediator versions agree, despite the 0.90 correlation between the two beliefs: controlling for the second-order belief alone absorbs 11.2 of the 15.0 pp (surveillance coefficient $-15.0 \rightarrow -3.8$ pp), against 6.1 for the first-order belief ($\rightarrow -8.9$ pp). Pre-decision elicitation rules out post-decision rationalization. Even so, this is a measured-mediator decomposition under a single manipulation—belief and action are both downstream of the treated messages—so it identifies the carrier descriptively rather than by intervention. The messages-included, pre-decision cell exists only for the primary model.

C Identification Test Details

Table A2 summarizes which candidate channels each test addresses and what the test concludes. The remaining tables in this section identify which text surface changes: briefing-body classifiers test whether the unchanged private briefing text can account for the action shift, while the peer-message classifier supports the message-stage degradation row.

Contamination and demand audit. Following Horton (2023), I test whether the models recognize the experiment. For the 4 accessible paper models, I issue 10 fresh post hoc calls per model per question. Each call presents the decision prompt and a $z = 0$ briefing without asking the model to play. Asked whether the setup resembles a formal model, 5% of responses name the global game or Morris-Shin specifically, and 78% identify a coordination

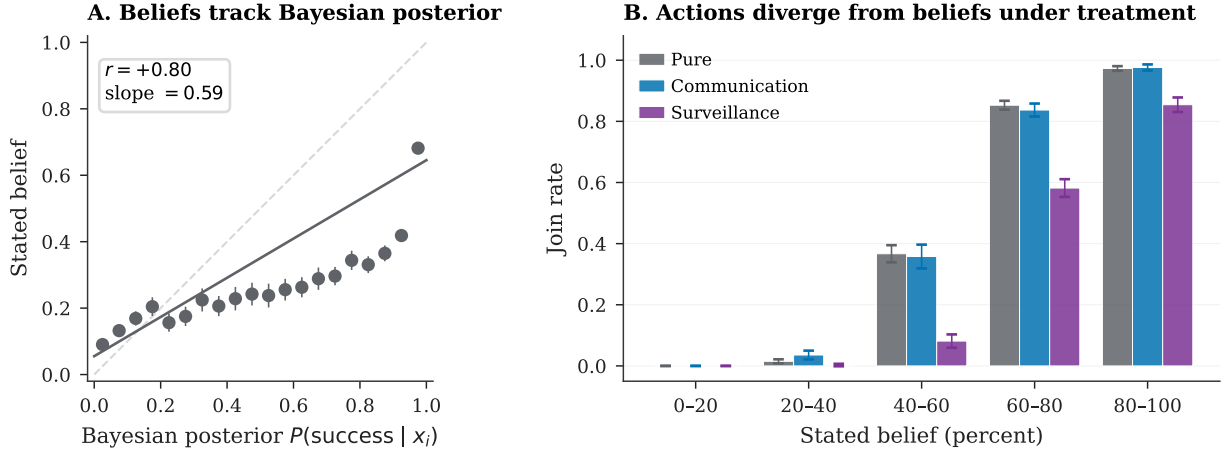


Figure A1: First-order belief elicitation (Mistral, private-information treatment, $N = 10,000$ agent-level observations). *Left*: Stated beliefs vs. the researcher-computed benchmark $P(\text{success} | x_i)$, binned means with 95% confidence intervals ($r = +0.80$). Agents are not shown the prior or signal noise. *Right*: Join rate by belief bin.

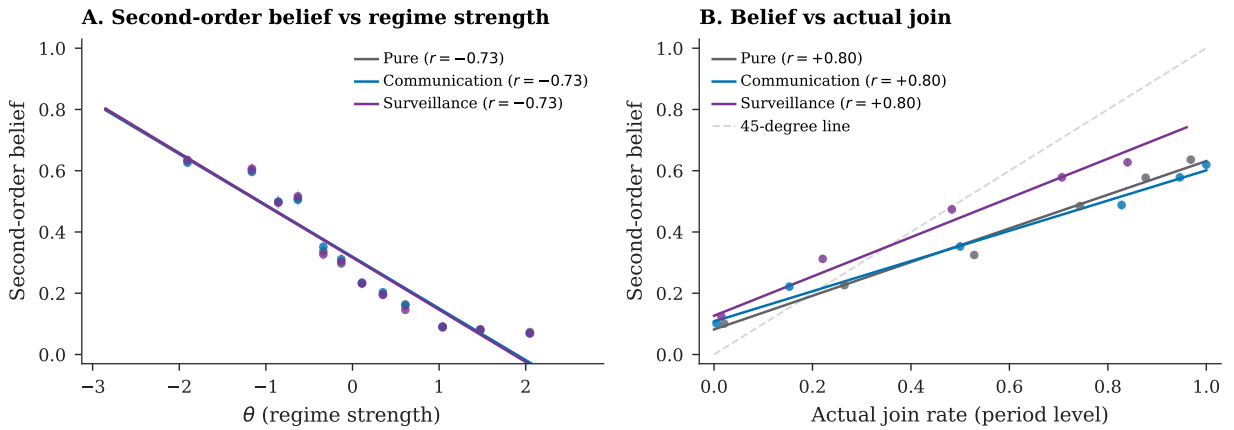


Figure A2: Second-order beliefs (Mistral, original primary belief run; restricted information set, messages excluded; binned means with 95% confidence intervals). *Panel A*: plotted against regime strength, the treatment arms largely overlap. *Panel B*: plotted against actual join rates, the surveillance curve sits higher because actions fall while restricted-information second-order reports move little. The factorial in Table A1 shows a significant shift when messages are included.

Table A1: Belief/action contrast under surveillance, primary model (Mistral Small Creative): effect of including peer messages in the belief elicitation prompt ($N = 12,500$ agent decisions per cell; 500 matched country-period cells across 10 countries, with country-clustered inference).

	Messages excluded		Messages included	
	Comm	Surv	Comm	Surv
First-order belief	46.0	45.7	53.2	47.4
Second-order belief	32.7	32.7	42.0	33.8
Join rate	42.4%	31.0%	42.2%	30.9%
<i>Surveillance effect (comm $\rightarrow$ surv):</i>				
Δ belief	−0.3 ($p = 0.002$)		−5.8 ($p = 0.002$)	
Δ SOB	0.0 ($p = 0.47$)		−8.2 ($p = 0.002$)	
Δ join	−11.4 pp ($p = 0.002$)		−11.3 pp ($p = 0.002$)	

Notes: Post-decision elicitation, 500 country-periods $\times$ 25 agents per cell ($N = 12,500$). “Messages excluded”: belief prompt sees briefing and decision only. “Messages included”: belief prompt additionally sees the peer messages received in the communication round. Beliefs and SOB are on the 0–100 scale; join rates in percent. The inferential unit is the country-period cell: each contrast is collapsed to the 500 matched country-period cell means and clustered on the 10 countries. Reported p -values are exact restricted wild-cluster bootstraps (full 2^{10} Rademacher enumeration over countries, so the minimum attainable p is $2/2^{10} \approx 0.002$); the cluster-robust t (9 dof) agrees throughout. Agent-level Welch tests that treat the 25 agents within a cell as independent give identical point estimates with mechanically smaller p -values and are reported in the replication output for comparison only. Deltas are differences of the displayed cell means.

or collective-action setting generically. Asked to predict the explicitly described surveillance effect, 82% of responses predict lower participation. The models can therefore recognize the hypothesis. The design does not require participant unawareness; the receiver-side estimate relies instead on excluding the warning from the decision prompt. Sender-side variants show that the response tracks monitoring attributed to the authorities, not the mere mention of monitoring: research-framed and anonymized monitoring leave joining essentially unchanged, while even a consequence-free routine-monitoring note attributed to the authorities reproduces most of the effect (Appendix D.3). The audit cannot be run on the retired primary model. Table A3 reports an action-prediction diagnostic; it is distinct from the baseline-versus-surveillance condition classifier. Classifiers trained on pure-treatment briefing text predict how agents would act from the private briefing alone. Applied to surveillance-treatment briefings, they predict join rates near the non-surveilled benchmark and miss the observed action drop. In contrast, a classifier on the actual peer messages—which are LLM-generated and downstream of the surveillance prompt—separates the two conditions sharply (91.9% accuracy on $N = 15,000$ held-out messages; Table A4). Together, the two exercises locate the manipulated text surface. The private briefing text misses the action shift; the surveilled peer messages change visibly.

D Robustness

Two kinds of checks bear directly on the surveillance effect: the falsification and placebo tests (Appendix D.2) and the warning phrasing and dose variants (Appendix D.3). The two full-scale checks of the same kind—the nested second-family replication and the cross-model writer–reader rotation—appear in the body (Section 5). The remaining subsections report aggregation accounting, agent-level regressions, and belief and text-baseline detail.

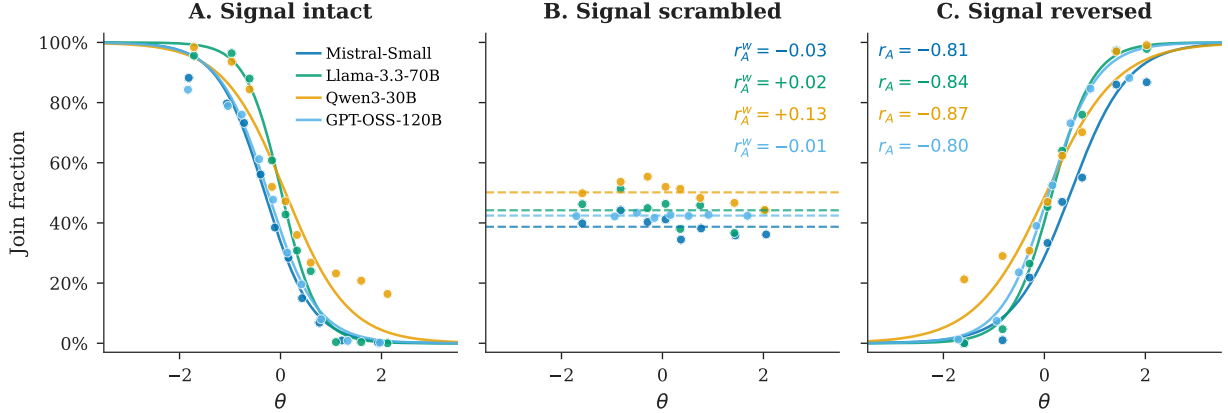


Figure A3: Falsification triptych for the behavioral benchmark (Section 4.1). Plotted series show four representative model families; reported mean correlations are across all 7 models in Table 3, with per-arm period counts in Table A12. *Left*: Pure (mean $r = +0.80$, raw per-model). *Center*: Scramble ($r = +0.03$, within-country). *Right*: Flip ($r = -0.80$).

D.1 Communication Aggregation

Communication aggregation accounting. Table A5 decomposes the communication estimators by model. The matched-cell estimand is the main one because it compares identical task cells. It keeps essentially all pure-treatment cells but drops communication-only cells in the larger communication arms. The gap relative to the pooled unpaired contrast is therefore driven mainly by repeated-cell weighting and arm-size differences—especially in the largest model—not by loss of pure-treatment support.

D.2 Falsification and Placebo Tests

Two clean placebo variants reframe the monitoring: “for research purposes” with no consequences, and “aggregated anonymously.” Neither deviates significantly from the communication baseline in the primary model (+0.3 pp, $p = 0.9351$, and +0.1 pp, $p = 0.9790$; Table A7). The Llama variants are small but imprecise. The decline is therefore tied to monitoring attributed to the regime in this prompt family, not to generic observation. Averaging the 500 matched-cell deltas within country gives $t = -13.90$ ($p < 0.001$).

D.3 Warning Phrasing and Dose Response

The clean surveillance treatment uses a single warning phrasing, so I vary that sentence directly. Two graded variants rerun the nested Llama 3.3 70B campaign on the identical 500-cell grid, changing only the appended warning. The *mild* variant reads “NOTE: Communications in your area are sometimes subject to routine monitoring by the authorities”; it removes the consequence language entirely. The *severe* variant reads “WARNING: Your communications ARE being monitored by regime security services. Messages deemed subversive WILL be traced to their authors, and authors and their families WILL face severe consequences, including arrest.” Joining falls by 6.8 pp under the mild warning ($t = -13.38$, $p < 0.001$, $N = 500$ matched cells) and by 10.0 pp under the severe warning ($t = -15.60$, $p < 0.001$). These bracket the 8.0 pp standard-warning effect on the same cells; the severe-minus-mild gap is -3.2 pp ($p < 0.001$). The response is monotone in stated severity, but most of it arrives at the lowest dose: the routine-monitoring note, which threatens nothing, already produces 85% of the standard-warning decline. The dose response rules out a one-sentence artifact. Within the nested campaign, most of the effect attaches to monitoring being attributed to the authorities at all, not to the rhetorical severity of the stated consequences. This matches the placebo pattern: monitoring framed as consequence-free research or anonymous aggregation produces no significant decline in the primary model (Table A7).

A sharper test changes the *format* of the monitoring cue rather than its wording. The **structural** variant conveys the same information as the standard warning but folds it into the scenario and channel description instead of appending a separate note. On the nested 500-cell grid it lowers joining by -1.5 pp (95% CI $[-1.9, -1.0]$, cluster-robust $t = -7.46$, wild $p < 0.01$, $N = 500$), compared with -8.0 pp under the appended warning. The sign survives; the magnitude is format-sensitive. These are effects of two specific prompt formats, not lower and upper bounds on an external monitoring effect.

Table A8 consolidates every Llama 3.3 70B surveillance estimate discussed in the paper, with its effect size, matched support, grid, and purpose. The table keeps the full-scale nested estimate (-8.0 pp, $N = 500$) and the noisy 20-cell prompt-isolation estimate (-15.0 pp) from being conflated.

D.4 Agent-Level Analysis

Table A9 reports agent-level logit regressions ($N = 240,557$, clustered SEs; per-column samples and clustering are stated in the table notes). The θ slope and the surveillance effect carry the expected signs. Some treatment indicators and interactions are not individually significant, so the table is descriptive rather than a complete hypothesis-test summary. A one-unit increase in θ reduces $P(\text{join})$ by 42 pp (derivative-based marginal effect at the mean). For the binary surveillance treatment, I report the average discrete-change effect: toggling the dummy and its θ -interaction jointly, then averaging over the sample, gives a 17.9 pp reduction in $P(\text{join})$. This has the same sign and order as the matched-cell contrast, though it is somewhat larger, because the regression pools arms and samples. (A derivative-at-the-mean calculation that treats the dummy as continuous and ignores the interaction would overstate the effect.) Column 3 estimates the association between elicited belief and action, controlling for the agent’s z -score. Beliefs in that column are elicited *post-decision*, so the near-perfect

pseudo- R^2 (0.975) likely reflects post-decision rationalization rather than a causal path from beliefs to action.

Figure A4(A) shows that a three-feature model (direction, clarity, coordination) modestly outperforms a one-feature sentiment baseline on average across the seven paper models, with several near-ties and one small negative gain; Panel (B) shows that the pure-trained predictor generalizes to communication but departs negatively under surveillance.

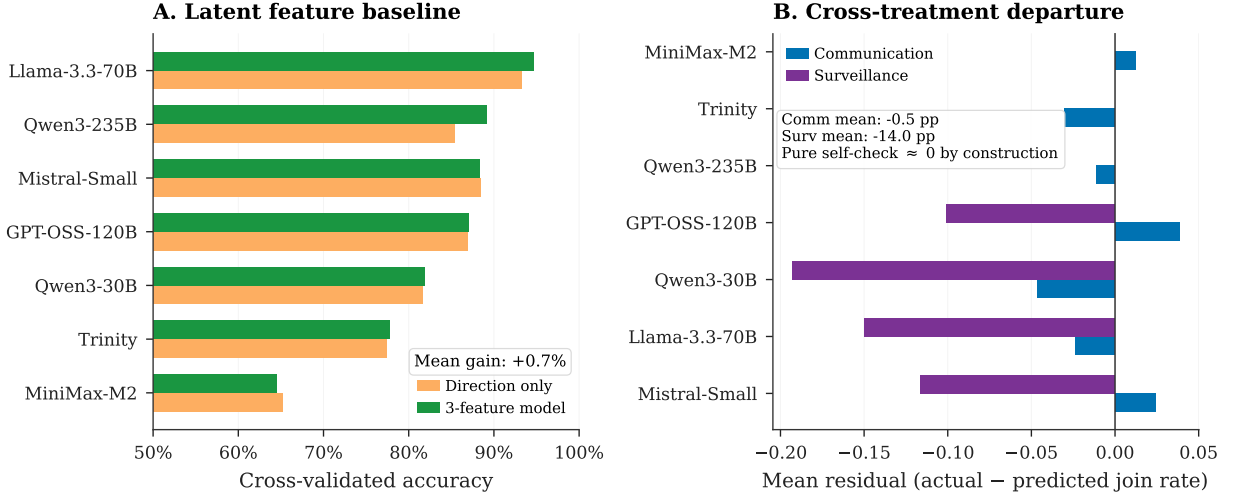


Figure A4: Construct validity tests. *Panel A*: cross-validated accuracy of a three-feature predictor (direction, clarity, coordination sliders) against a one-feature sentiment baseline, per model. *Panel B*: mean residual (actual – predicted join rate) when the pure-trained predictor is applied to the communication and surveillance arms; the pure-treatment self-check is ≈ 0 by construction and omitted. The predictor generalizes to communication but departs negatively under surveillance.

D.5 Belief Alignment

Stated beliefs track the researcher-computed benchmark success probability ($r_{\text{belief,posterior}} = +0.80$) and predict actions ($r_{\text{belief,decision}} = +0.84$; partial $r = +0.64$ controlling for signal; Table A11). Agents are not shown the prior or noise parameters behind that benchmark.

D.6 Cross-Model and Text Baseline Details

Figure A5 plots per-model state-response correlations under the private-information and communication treatments. Figure A6 compares the LLM join curve with a naïve text-sentiment predictor built directly from the briefing generator’s direction slider; the predictor co-moves with observed joining ($r = 0.80$), as expected since briefing content derives from the same slider.

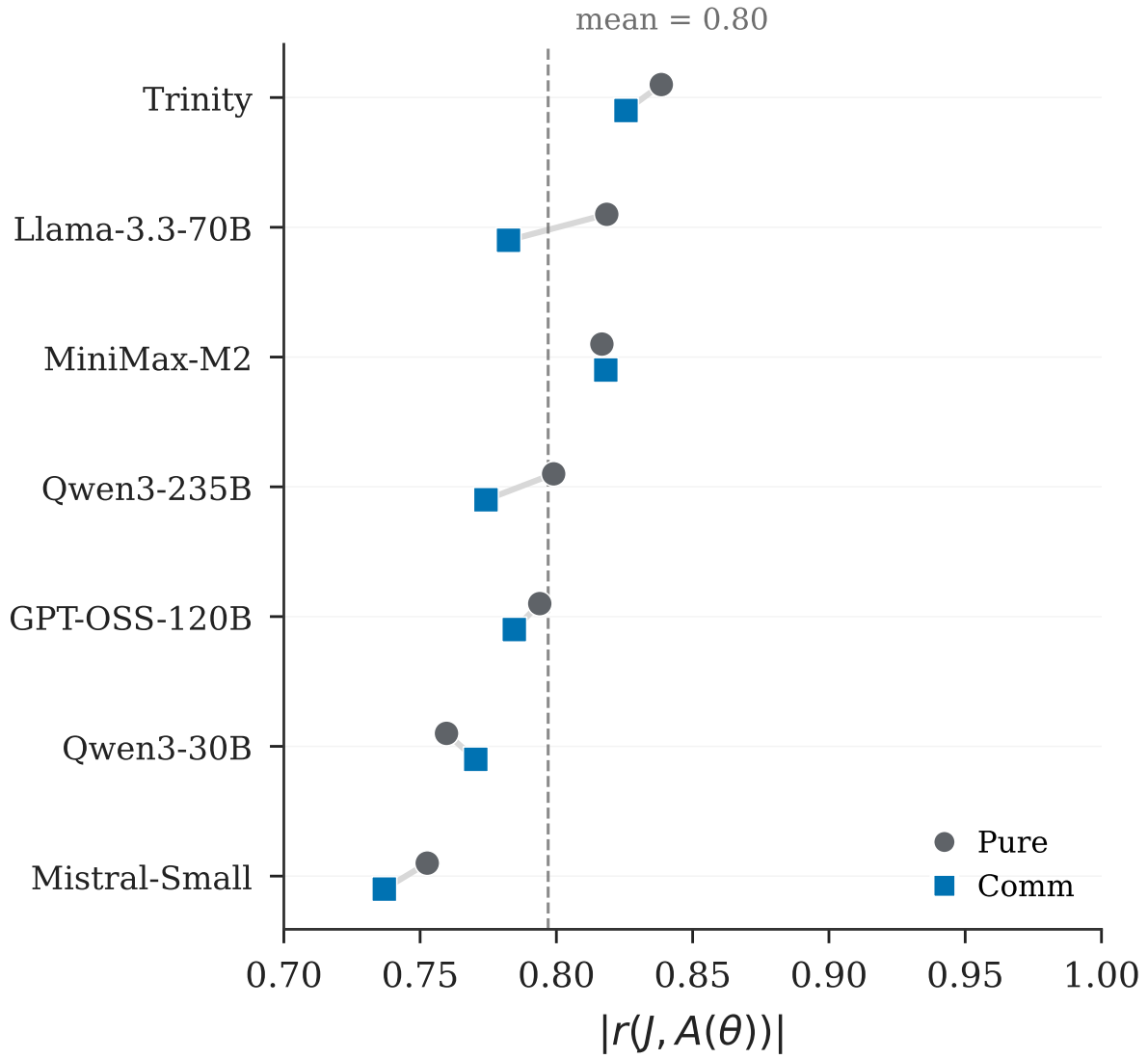


Figure A5: Cross-model signal monotonicity. Points report $|r(J, A(\theta))|$ under private-information and communication treatments for each model; near-identical pairs are vertically offset for visibility.

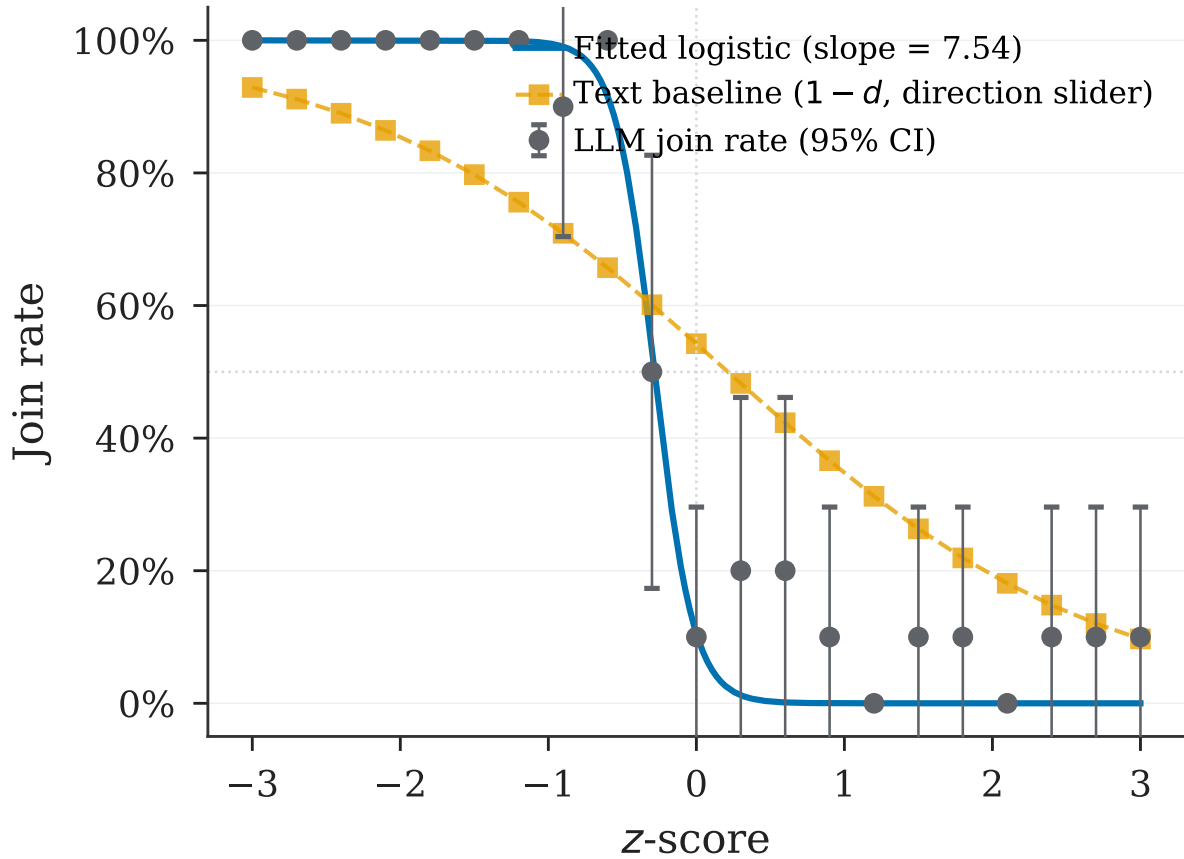


Figure A6: Text baseline test (Mistral, private-information treatment). The LLM join curve is compared with the naïve text predictor $1 - d$, where d is the briefing-generator direction slider. Error bars are 95% confidence intervals on binned join rates.

E Implementation Details

Unless otherwise stated, LLM calls use temperature = 0.7, `max_tokens` = 512, single sample per decision, routed through OpenRouter. Model identifiers are listed in Table 2; exact request payloads and responses are cached by hash because hosted model snapshots may change over time.⁵

Before the core runs, a calibration sweep varies only `cutoff_center`, the constant subtracted from the raw briefing z -score before the direction, clarity, and coordination sliders are computed. The selection objective penalizes fitted-midpoint error, curve-fit error, and API failures; it never targets the join-curve slope. Table 2 reports the logged value for every paper model. Because centers differ, the same raw z can generate different prose across models. All arms compared within a model share the same center, so treatment contrasts hold this map fixed. Cross-model response levels should therefore not be read as responses to identical prose.

Each country has a researcher-side state mean $\bar{z}_c \sim \mathcal{N}(0, 0.3^2)$. Each period draws a perturbation $\eta_{ct} \sim \mathcal{N}(0, 0.05^2)$ and sets $z_{ct} = \bar{z}_c + \eta_{ct}$. Regime strength is then drawn as $\theta_{ct} \sim \mathcal{N}(z_{ct}, 1)$, and private signals are $x_i = \theta_{ct} + \varepsilon_i$ with $\sigma = 0.3$. These distributions generate the experiment cells; they are not shown to agents as explicit priors. The communication network is Watts–Strogatz ($k = 4$, $p = 0.3$), constructed once per run from the master seed and reused across period-level task rows; agents are reinitialized in each row. All random draws use a master seed logged in per-run JSON manifests, and LLM responses are cached by request hash for exact reproducibility.

Responses are parsed for JOIN/STAY tokens. Combined error rates are below 2% in every listed treatment for 5 of 7 models. There are two exceptions. GPT-OSS 120B has unparseable-response rates of 3.1–3.3%, and only in the scramble and flip falsification arms. Trinity Large has elevated API errors ($\approx 9\%$) caused by provider content filtering, so its missingness remains a model-specific limitation. All statistics use `join_fraction_valid` (Table A12).

The nested revision campaign (Section 5) runs on Llama 3.3 70B with an identical grid (10 countries $\times$ 50 periods, $n = 25$, seed 5150, temperature 0.7). Its arms are a communication baseline, a clean monitoring arm, a codedness-induction control, graded warning arms, and raw/decoded message replays. For Experiments A and B, Qwen3 30B rewrites the source messages, and fresh Llama receivers replay them in exact fixed-message mode on the same 500-cell grid, with seed 4242 and `decision_context=none`. Experiment A crosses explicit willingness content with direct versus coded style. Experiment B uses baseline, monitoring, risk-only, and risk-stripped messages. Inference collapses decisions to matched cells and enumerates all 2^{10} country-cluster sign assignments. The replay point estimates use valid JOIN/STAY decisions. Valid rates are 82.9%–83.5% in Experiment A and 87.4%–94.0% in Experiment B; Table 6 reports the uniform all-JOIN/all-STAY sensitivities. The committed cell-level CSV and the generator command are documented in the data manifest.

The dissent floor parameter (default 0.25) sets the minimum probability of contrary-valence

⁵The primary model identifier `mistralai/mistral-small-creative` was retired from OpenRouter after data collection. All cached request/response pairs used in this paper are archived in the replication package, but new experiments on that exact endpoint cannot be issued; replicators wishing to extend the primary-model arm should consult the data archive rather than re-querying the live API.

cues in any briefing. With the logistic direction slider (slope = 0.8), the valence-channel separation between extreme signals ($z = \pm 2$) is $0.664 \cdot (1 - 2 \cdot \text{floor})$. At the default 0.25, the valence channel retains 50% of its floor-free maximum; halving the floor to 0.125 retains 75%; at 0.50 the valence channel collapses to zero by construction. The clarity and coordination sliders depend on z independently of the floor and carry signal at all floor values, so the threshold relation does not hinge on this design choice.

Code, prompt templates, logged model outputs, analysis data, replication instructions, and asset lineage are available at <https://github.com/keltokhy/llm-global-games>.

E.1 Briefing Generator Examples

$z = -2$ (weak regime)	$z = 0$ (ambiguous)	$z = +2$ (strong regime)
PRIVATE BRIEFING BOTTOM LINE: What you're seeing is not a rough patch – it's a pattern of accelerating deterioration. ATMOSPHERE: The fear barrier has collapsed. People who were terrified last week are now openly discussing action. EVIDENCE: - [Elite Cohesion] a once-loyal advisor has been publicly contradicting official statements (near-unanimous confirmation from every reliable source) - [Security Forces] rank-and-file soldiers are grumbling about pay delays and broken promises (overwhelming convergence – every channel telling the same story independently) - [Money And Logistics] banks are imposing new withdrawal limits, citing 'temporary liquidity management' (overwhelming convergence – every channel telling the same story independently) - [Street Mood] street vendors have returned to areas they'd avoided during previous periods of tension – the all-clear has been internalized (overwhelming convergence – every channel telling the same story independently) - [Information Control] the regime's daily press briefing has been cancelled three times this week – they can't decide what to say - [Personal Observations] people you trust are making quiet preparations – nothing dramatic, but noticeable (overwhelming convergence – every channel telling the same story independently) - [Diplomatic Signals] international shipping companies have reclassified the country as higher-risk for cargo insurance (overwhelming convergence – every channel telling the same story independently) - [Institutional Functioning] civil servants are reporting contradictory instructions from different parts of the bureaucracy (overwhelming convergence – every channel telling the same story independently) UNCERTAINTIES: - How deep the fractures go – surface-level dysfunction can mask either deeper rot or functioning parallel systems - Whether the military's silence means they're loyal, undecided, or waiting for the right moment.	PRIVATE BRIEFING BOTTOM LINE: The situation is balanced on a knife's edge – small events could determine which way it falls ATMOSPHERE: Trusted intermediaries are moving between groups, relaying assessments and gauging willingness. EVIDENCE: - [Elite Cohesion] elite cohesion is strained but not broken – rivalries are being managed rather than resolved - [Security Forces] the security forces are maintaining presence but seem to be conserving resources rather than demonstrating strength - [Money And Logistics] the currency is under pressure but hasn't cracked – traders are nervous but still trading - [Street Mood] younger people are visibly more restless than older people, who counsel patience and caution - [Information Control] the regime's information apparatus looks intact but tired – producing content without the energy or creativity it once had (a couple of contacts mentioning it independently) - [Personal Observations] people seem watchful rather than panicked – paying closer attention but not acting differently - [Diplomatic Signals] a regional power issued a carefully worded statement supporting 'institutional stability' – ambiguous enough to cover any outcome - [Institutional Functioning] routine government business continues but with more errors and corrections than usual (a couple of contacts mentioning it independently) UNCERTAINTIES: - If external conditions change – an economic shock, a regional crisis – whether the regime's stability would hold. - Whether the regime's recent successes have made it complacent about emerging threats it should be managing.	PRIVATE BRIEFING BOTTOM LINE: By every measure available to you, the regime's hold on power is firm and likely to remain so ATMOSPHERE: Small gestures of solidarity happen – a knowing look, a shared silence – but nothing that could be called coordination. EVIDENCE: - [Elite Cohesion] a former rival of the leader has been publicly praising the regime's direction (near-unanimous confirmation from every reliable source) - [Security Forces] a contact in the barracks says morale is solid and orders are flowing cleanly (overwhelming convergence – every channel telling the same story independently) - [Money And Logistics] a major government infrastructure contract was just awarded – business as usual (overwhelming convergence – every channel telling the same story independently) - [Street Mood] street vendors have returned to areas they'd avoided during previous periods of tension – the all-clear has been internalized (overwhelming convergence – every channel telling the same story independently) - [Information Control] an opposition figure's leaked criticism was preemptively addressed before it gained traction – the intelligence apparatus is ahead of events - [Personal Observations] a well-connected friend said the situation is 'messy but controlled' and seemed unbothered (overwhelming convergence – every channel telling the same story independently) - [Diplomatic Signals] foreign students continue to enroll at local universities – their embassies haven't advised them to leave (overwhelming convergence – every channel telling the same story independently) - [Institutional Functioning] civil servants are reporting contradictory instructions from different parts of the bureaucracy (overwhelming convergence – every channel telling the same story independently) UNCERTAINTIES: - If the regime's information advantage is as complete as it appears, or if opponents are communicating through channels you can't see. - Whether any latent opposition exists that simply hasn't shown itself yet – absence of evidence is not evidence of absence.

Figure A7: Example briefings at three centered z -scores. *Left:* $z = -2$ —weak-regime cues, open coordination language, and high clarity. *Center:* $z = 0$ —mixed cues and low clarity. *Right:* $z = +2$ —strong-regime cues, guarded coordination language, and high clarity. The displayed text is an unedited generator draw at seed 5150, agent 2, period 1007; the generator retains some contrary-valence cues through the dissent floor.

Table A13 reports the three slider values at representative *centered* inputs z – *cutoff_center*. The direction slider is logistic with slope 0.8. Clarity is U-shaped around zero, and coordination decreases with the centered input. These are generator-function values, not the raw- z mapping used by every model; the model-specific centers in Table 2 shift that mapping before each core run.

Figure A7 shows complete briefings generated at $z \in \{-2, 0, +2\}$. At the extremes ($|z| = 2$), clarity is high (0.98) and the cue balance is strongly directional. At $z = 0$, clarity is 0.00 (maximally ambiguous) and the briefing presents a balanced picture.

Table A2: Identification map. Each row lists a candidate channel through which the surveillance treatment could produce the observed action effect; the right column identifies the test(s) addressing it and the conclusion. “Inconclusive” marks channels left bundled by the current design.

Candidate channel	Test & outcome
<i>Direct warning exposure at decision time</i>	The main decision prompt contains no warning (<code>decision.context=none</code>). A hold-fixed test shows that adding the warning there moves joining by -20.9 pp. Direct exposure is absent from the main treatment, but receivers may infer risk from generated messages.
<i>Generic observation effect</i>	Placebo (“monitoring for research, no consequences”) and anonymous monitoring variants (Appendix D.2). Both are null in the primary model ($p > 0.9$); Llama variants are small but imprecise. <i>Disfavored as the main driver.</i>
<i>Coded form</i>	A codedness-induction control without monitoring framing moves joining by -0.7 pp, against -8.0 pp under monitoring on the same cells. In the 2×2 replay, direct form raises joining by +4.30 or +1.86 pp with willingness held fixed. <i>Form matters but is insufficient.</i>
<i>Willingness/actionability content</i>	The fixed-message 2×2 replay holds style and factual assessment fixed. Stating that others are willing to act raises parseable joining +7.94 pp in direct form and +5.50 pp in coded form. Valid-response rates are balanced, and uniform all-JOIN/all-STAY recodings preserve the sign. <i>Strongest experimentally manipulated pathway.</i>
<i>Repression-risk content</i>	Adding one risk clause to baseline messages lowers parseable joining -3.99 pp, while risk-stripped monitored messages return near baseline. Valid-response rates differ across arms, the stripped rewrite also sharpens actionability, and an all-JOIN recoding erases the risk-only contrast. <i>Suggestive, not separately resolved.</i>
<i>Same-family reflexivity</i>	Bidirectional cross-model rotation is significant in both directions: Llama writes $\rightarrow$ Qwen3 reads (-7.5 pp) and Qwen3 writes $\rightarrow$ Llama reads (-1.9 pp). Same-family reflexivity is insufficient, although the cross-direction asymmetry remains unexplained.

Table A3: Classifier baselines. Accuracy and AUC are 5-fold CV on pure-treatment data. “Pred. join (surv.)” is the classifier’s predicted join rate when applied to surveillance-treatment briefings; “Actual” is the LLM’s observed rate. The gap measures the surveillance action shift invisible to briefing-body text classifiers.

Classifier	Acc.	AUC	Pred. (surv.)	Actual	Gap
BoW TF-IDF	88.5%	0.947	40.6%	27.6%	+13.0 pp
Slider logistic	88.4%	0.941	40.4%	27.6%	+12.8 pp
Keyphrase	60.2%	0.594	8.4%	27.6%	−19.2 pp

Notes: Primary model (Mistral Small Creative). Accuracy and AUC from 5-fold cross-validation on pure-treatment data. Training sample: the primary model’s pure arm (1,000 country–period rows, 25 agents per row, 25,000 agent decisions); each CV fold trains on 80% and tests on the held-out 20%. The surveillance application scores all 1,000 surveillance-arm rows (25,000 agent decisions). Gap = classifier-predicted join rate − actual LLM join rate under surveillance (pp).

Table A4: Top 10 most discriminative uni- and bi-grams between baseline and surveillance peer messages (primary model). A logistic classifier on tf-idf features achieves 91.9% accuracy on a held-out sample of $N = 15,000$.

Surveillance-distinguishing		Baseline-distinguishing	
Feature	Coef	Feature	Coef
things feel	+6.41	look the	−11.11
walls are	+5.87	look	−8.93
are cracking	+5.64	listen	−7.31
things	+5.56	listen this	−6.00
cracking even	+4.96	may	−4.93
cracking	+4.46	isn just	−4.68
ground shifting	+3.79	don panic	−4.63
the walls	+3.54	the weather	−4.49
feel like	+3.50	weather	−4.38
rush this	+3.37	something shifting	−4.26

Notes: Logistic regression on tf-idf uni- and bi-grams (max 5,000 features, min document frequency 10). Train/test split 70/30, stratified ($N_{\text{train}} = 35,000$, $N_{\text{test}} = 15,000$). Positive coefficients distinguish surveillance messages, negative coefficients distinguish baseline. The sample contains only baseline communication and clean surveillance peer messages; all other conditions are excluded. Direct-reference and coded-metaphor claims in the text come from separate dictionary counts, not from this top-n-gram list alone.

Table A5: Communication estimators: aggregation accounting.

Model	Rows/arm	Row wt. (P/C)	Cells/arm	Matched	Cell wt.	Lost (P/C)	Δ unpaired (pp)	Δ paired (pp)
Mistral Small Creative	1000	55.6%/38.5%	680	680	46.0%	0/0	+2.38	+4.84
Llama 3.3 70B	100	5.6%/3.8%	100	100	6.8%	0/0	-2.36	-2.36
Qwen3 30B	100	5.6%/3.8%	100	100	6.8%	0/0	-4.64	-4.64
GPT-OSS 120B	200	11.1%/7.7%	200	200	13.5%	0/0	+3.47	+3.47
Qwen3 235B	200/1000	11.1%/38.5%	200/1000	200	13.5%	0/800	+0.86	+0.12
Trinity Large	100/99	5.6%/3.8%	100/99	99	6.7%	1/0	-3.14	-3.41
MiniMax M2-Her	100	5.6%/3.8%	100	100	6.8%	0/0	+1.32	+1.32
Equal-weight avg	—	14.3% each	—	—	14.3% each	—	-0.30	-0.09
Pooled	1800/2599	100.0%	1480/2279	1479	100.0%	1/800	+1.39	+2.10

Notes: Sample: valid country-period rows from the core pure and communication suites, each aggregating $n = 25$ agent decisions. “Rows/arm” counts valid country-period rows entering the pooled unpaired estimator in each arm. “Cells/arm” counts unique task cells after collapsing duplicates by the exact matching key (model, country, period, θ , z , benefit, θ^*). The paired estimator averages within these cells and keeps only common support; “Row wt.” and “Cell wt.” are the model shares in the pooled unpaired and pooled paired estimators, respectively. The pooled paired Δ of +2.10 pp has a paired standard error of 0.33 pp ($t = 6.38$, $p < 0.001$, $N = 1,479$ matched cells). Trinity Large has one pure-only cell. Qwen3 235B has 800 communication-only cells because its communication arm is larger than its pure arm; those cells are excluded from the paired common-support estimator.

Table A6: Clean prompt-isolation surveillance reruns. The surveillance prompt is the baseline trusted-contact communication prompt plus one appended monitoring warning; the decision prompt is unchanged.

Model	Full cells (B/S)	Matched cells	Baseline	Surv.	Δ (pp)	SE	$r(J, A)$	p
Mistral Small Creative	680/1000	500	45.3%	31.4%	-13.9	0.7	+0.72	<0.001
Llama 3.3 70B	100/1000	20	46.2%	31.2%	-15.0	4.4	+0.65	0.003
Qwen3 30B	100/1000	20	50.0%	37.8%	-12.2	2.5	+0.68	<0.001
GPT-OSS 120B	200/1000	20	42.5%	30.5%	-12.0	3.9	+0.75	0.006
Qwen3 235B	1000/1000	20	42.0%	39.0%	-3.0	2.2	+0.78	0.186
Equal-weight avg	—	—	—	—	-11.2	—	—	—
Pooled matched	—	580	—	—	-13.5	0.7	—	<0.001

Notes: Baseline is each model’s communication treatment. “Full cells” counts the available baseline and monitoring task cells after collapsing duplicate rows by the exact key (country, period, θ , z , benefit, θ^*); it is not the support used for the paired effect when common support is smaller. The non-primary baseline communication arms were shorter pilot grids (100–200 cells) while the later monitoring reruns used a larger grid; because those grids were not nested, only 20 exact cells overlap for several models. N for each paired contrast is the number of matched common-support cells; Baseline, Surv., Δ , SE, and p are computed only on those cells. $r(J, A)$ is the monitoring-arm correlation with the external attack-mass benchmark on the full rerun. Aggregating the primary model’s matched-cell deltas to country means ($N = 5$ countries) gives $t = -13.90$, $p < 0.001$, so the headline effect survives country-level clustering.

Table A7: Surveillance isolation checks. Placebo: monitored for research, no consequences. Anonymous: messages aggregated anonymously. Neither deviates significantly from the communication baseline.

Model	Variant	N	Mean join	$r(J, A)$	Δ (pp)	p
Mistral	Placebo	200	41.4%	+0.73	+0.3	0.935
	Anonymous	200	41.2%	+0.73	+0.1	0.979
Llama 3.3 70B	Placebo	1000	41.2%	+0.71	-0.3	0.940
	Anonymous	1000	40.5%	+0.72	-1.1	0.814

Notes: Placebo: agents told monitoring is for research purposes only (no consequences). Anonymous: messages aggregated before delivery. N counts country-period rows ($n = 25$ agents per row). Δ : change vs. the same model’s communication baseline (pp), with p from a two-sample t -test against that baseline. Communication-baseline arms: Mistral: $N = 1000$, mean join 41.1%; Llama 3.3 70B: $N = 100$, mean join 41.6%.

Table A8: Every Llama 3.3 70B surveillance estimate: effect size, matched support, grid, and purpose. All deltas are surveillance (or the stated arm) minus its own baseline on the matched cells. The full-scale nested estimate (-8.0 pp, $N = 500$) is the headline cross-model confirmation; the original prompt-isolation estimate (-15.0 pp) rests on only 20 overlapping cells and is a noisy sign check, not a full replication.

Arm	Δ (pp)	Matched N	Grid	Purpose
<i>Surveillance (monitoring warning), coordination task:</i>				
Nested clean surveillance	-8.0	500	nested 10×50 , common seed	full-scale confirmation (§5)
Prompt-isolation (original)	-15.0	20	20-cell pilot/rerun overlap	original sign check (Table A6)
Mild routine-monitoring note	-6.8	500	nested 10×50	dose response (Appendix D.3)
Severe-consequence note	-10.0	500	nested 10×50	dose response (Appendix D.3)
<i>Replay / decoding of surveilled messages, coordination task:</i>				
Raw-message replay	-7.9	500	nested 10×50	validates replay design ($\approx$ live)
Decoded-message replay	-4.9	500	nested 10×50	decoding restores participation
<i>Cross-model writer-reader rotation:</i>				
Within-Llama (matched baseline)	-6.0	200	rotation cells	reflexivity baseline (§5)
Llama writes, Qwen3 30B reads	-7.5	200	rotation cells	cross-family transmission
Qwen3 30B writes, Llama reads	-1.9	200	rotation cells	cross-family transmission
<i>Other decision task:</i>				
Private bet on regime fall	-5.6	500	cross-task grid	task contrast (§5)
<i>Reference control (no monitoring framing):</i>				
Codedness-induction control	-0.7	500	nested 10×50	isolates coded form alone

Notes: “Matched N ” is the number of common-support cells entering each paired contrast. “Nested” arms share the identical 500-cell grid (10 countries $\times$ 50 periods, $n = 25$, common seed) and are matched by construction. The original prompt-isolation estimate shares only 20 cells with its pilot baseline because those grids were not nested. Rotation contrasts are matched on (country, period, θ) cells across writer and reader models. The private-bet and nested-coordination samples are not cell-matched to each other.

Table A9: Agent-level regressions

	(1) Join Decision Logit		(2) Coord. Ablation Logit		(3) Belief → Action Logit	
θ	−1.745***	(0.043)				
Direction			−14.088***	(0.996)		
Coordination			−9.333***	(1.131)		
Dir × Coord			−0.042	(0.360)		
Belief					0.353***	(0.023)
z -score					−0.026	(0.082)
Comm	0.017	(0.023)				
Flip	−0.007	(0.055)				
Scramble	0.068*	(0.038)				
Surveillance	−1.596***	(0.048)				
$\theta \times$ Comm	−0.538***	(0.037)				
$\theta \times$ Flip	3.308***	(0.070)				
$\theta \times$ Scramble	1.702***	(0.044)				
$\theta \times$ Surveillance	−1.151***	(0.062)				
Treat: Pure					4.673***	(0.981)
Treat: Surveillance					0.021	(0.244)
Constant	−0.086*	(0.051)	11.022***	(1.065)	−25.429***	(1.686)
Model FE	Yes		No		No	
Clustered SE	Yes		Yes		Yes	
N	240,557		44,662		14,990	
Pseudo R^2	0.353		0.442		0.975	

Notes: Logit coefficients with standard errors clustered at the model–country–period level in parentheses (all three columns; in column (3) the sample is almost entirely a single model, so clusters effectively coincide with country–period cells). Column (1): agent-level join decisions pooled across all seven models in the pure, communication, scramble, and flip treatments, plus surveillance runs for Mistral, Llama 3.3 70B, and Qwen3 30B ($N = 240,557$; 4,700 clusters). Regressors: θ , treatment dummies (base category: pure), $\theta \times$ treatment interactions, and model fixed effects. Column (2): coordination ablation— $\Pr(\text{join}_i = 1) = \Lambda(\beta_0 + \beta_1 \text{Dir}_i + \beta_2 \text{Coord}_i + \beta_3 \text{Dir}_i \times \text{Coord}_i)$, where Λ denotes the logistic CDF—using briefing slider values; pure treatment only, all seven models pooled, no model fixed effects ($N = 44,662$; 1,400 clusters). Direction $\in [0, 1]$ is increasing in regime strength; Coordination $\in [0, 1]$ is decreasing (high = weaker regime). Because both are monotone logistic transforms of the same signal, they are near-collinear ($\rho \approx -1$); individual coefficients reflect the partial effect *conditional on the other*, not the marginal effect. Column (3): partial effect of elicited belief ($b_i \in [0, 100]$, stated success probability) on action, controlling for z -score. Sample: belief-elicitation runs only (communication, pure, and surveillance variants), comprising Mistral (14,981 obs.), Llama 3.3 70B (9 obs.); $N = 14,990$; 369 clusters. “Treat: X” rows are treatment intercept dummies (base category: communication), not interactions with belief. Marginal effects, column (1): for the continuous regressor θ , the derivative-based effect at the mean is $\hat{\beta}_\theta \bar{p}(1 - \bar{p}) = -0.416$. For the binary surveillance treatment, a derivative-based marginal effect is not meaningful and would ignore the $\theta \times$ surveillance interaction; we therefore report the average discrete-change effect of surveillance (toggling the dummy and its θ -interaction jointly, with all other regressors at observed values, averaged over the column (1) estimation sample): −17.9 pp. **Caveat:** beliefs are elicited *after* the join/stay decision (post-decision), so the belief coefficient reflects association rather than a causal pathway; post-decision rationalization may inflate the correlation. The very high pseudo- R^2 (0.975) and large intercept indicate near-deterministic prediction consistent with quasi-separation—belief almost perfectly predicts action, as expected when belief is stated *post hoc*. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Logistic fit parameters by model and treatment. $\hat{\theta}^*$ is the estimated cutoff ($-b_0/b_1$); $\kappa \equiv b_1$ is the steepness parameter.

Model	Pure			Communication		
	$\hat{\theta}^*$ (SE)	κ (SE)	N	$\hat{\theta}^*$ (SE)	κ (SE)	N
Mistral Small Creative	-0.32 (0.02)	+2.15 (0.08)	1000	-0.23 (0.02)	+2.57 (0.11)	1000
Llama 3.3 70B	+0.02 (0.04)	+2.83 (0.36)	100	-0.06 (0.04)	+3.63 (0.52)	100
Qwen3 30B	+0.10 (0.06)	+1.62 (0.16)	100	-0.03 (0.04)	+2.94 (0.36)	100
GPT-OSS 120B	-0.25 (0.04)	+2.06 (0.15)	200	-0.15 (0.03)	+3.03 (0.26)	200
Qwen3 235B	-0.22 (0.03)	+2.11 (0.15)	200	-0.18 (0.01)	+2.78 (0.10)	1000
Trinity Large	+0.08 (0.05)	+1.34 (0.11)	100	-0.02 (0.04)	+2.23 (0.23)	100
MiniMax M2-Her	-0.17 (0.09)	+0.61 (0.06)	100	-0.07 (0.09)	+0.66 (0.06)	100

Notes: Fitted form: $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1 \theta})$. $\hat{\theta}^* = -b_0/b_1$: estimated cutoff. $\kappa \equiv b_1$: logistic steepness (larger = sharper threshold). Sign convention: positive κ corresponds to a join curve decreasing in θ ; in the agent-level logit (Table A9) the same comparative static appears as a negative coefficient on θ . N counts period-level rows entering each fit. Standard errors from the covariance matrix of the nonlinear fit; cutoff SE by delta method.

Table A11: Agent-level belief correlations (primary model: Mistral Small Creative).

	N	$r_{\text{belief,posterior}}$	$r_{\text{belief,decision}}$	r_{partial}	Mean belief (%)
Pure	10,000	+0.80 [0.79, 0.80]	+0.84 [0.83, 0.85]	+0.64	44.4
Comm.	5,000	+0.80 [0.79, 0.81]	+0.83 [0.82, 0.84]	+0.56	44.4
Surv.	5,000	+0.79 [0.78, 0.80]	+0.73 [0.72, 0.74]	+0.47	43.6

Notes: N counts agent-level observations (one elicited belief per agent decision). Beliefs are elicited on a 0–100 probability scale. $r_{\text{belief,posterior}}$ compares stated belief with the researcher-computed benchmark $P(\theta < \theta^* \mid x_i)$; agents are not shown the prior or noise parameters needed to compute it. $r_{\text{belief,decision}}$ compares belief with the JOIN/STAY decision. r_{partial} controls for the signal z-score.

Table A12: Parse error and API failure rates by model and treatment.

Model	Treat.	N	API err	Unparse.	Combined
Mistral	Pure	1000	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	200	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Llama 3.3 70B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Qwen3 30B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
GPT-OSS 120B	Pure	200	0.0%	1.5%	1.5%
	Comm	200	0.0%	0.3%	0.3%
	Scr.	500	0.0%	3.1%	3.1%
	Flip	500	0.0%	3.3%	3.3%
Qwen3 235B	Pure	200	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Trinity	Pure	100	9.0%	0.0%	9.0%
	Comm	100	10.0%	0.0%	10.0%
	Scr.	100	8.0%	0.0%	8.0%
	Flip	100	9.9%	0.0%	9.9%
MiniMax M2	Pure	100	0.0%	1.5%	1.5%
	Comm	100	0.0%	0.9%	0.9%
	Scr.	100	0.0%	1.8%	1.8%
	Flip	100	0.0%	1.0%	1.0%

Notes: Per-period averages. API error = provider-side failure (timeout, rate limit, content filter). Unparseable = valid response not classified as JOIN/STAY. Combined < 2% in every listed treatment for five of seven models. The two exceptions are GPT-OSS 120B, with 3.1–3.3% unparseable responses in the scramble and flip arms, and Trinity, with 8–10% API errors from provider content filtering. Trinity’s elevated API errors ($\approx 9\%$) mean its missingness is reported rather than resolved. A signal-balance diagnostic for Trinity’s per-period API-error rate gives $\max |r(\theta, \text{error})| = 0.19$ over the listed treatments (minimum $p = 0.064$).

Table A13: Slider values at representative centered inputs $z - \text{cutoff_center}$.

z	Direction	Clarity	Coordination
−2.0	0.17	0.98	0.77
−1.0	0.31	0.63	0.65
0.0	0.50	0.00	0.50
+1.0	0.69	0.63	0.35
+2.0	0.83	0.98	0.23